

Acknowledgements

[TODO: write at the end]

Contents

Acknowledgements	1
1 Introduction	4
2 Literature review	8
2.1 From text foundation models to single-cell foundation models	8
2.2 Cell sentences, and the model this thesis fine-tunes	8
2.3 The dispute: do these models beat uninformative baselines?	10
2.4 Six ways a drug-response benchmark can mislead	11
2.5 Connectivity mapping and signature transfer	12
2.6 Decodable, used, and the global workspace	12
2.7 Optimal transport for state matching	13
3 Methods	16
3.1 Data	16
3.2 Model and representation	17
3.3 Metrics	18
3.4 Controls	20
3.5 Mechanistic probing	25
3.6 Target constructions	25
3.7 Residual targets	26
3.8 Generalisation design	27
3.9 Baselines	28
3.10 Variance decomposition	29
4 Results and Analysis	31
4.1 The metric is the first result	31
4.2 The model is drug-blind	32
4.3 Where the blindness lives	36
4.4 The ε -ladder: no target-side fix works	39
4.5 The repair: it was the tokenisation	40
4.6 The drug use generalises	42
4.7 But a lookup table wins	43
4.8 Every arm in one place	44
4.9 How much is left? The transfer coefficient	44
4.10 Scope: what is closed by measurement	47
4.11 But the model does not read the channel it already has	48

5	Limitations and Future Research Directions	50
5.1	Limitations of scope	50
5.2	Limitations of the unseen-drug evaluation	51
5.3	Directions	51
6	Conclusions	55
A	Appendix	62
A.1	Estimator validation against planted ground truth	62
A.2	Absent-gene convention	62
A.3	Swap-distance sweep	62
A.4	Training configuration	63
A.5	Calibration self-test	64

1 Introduction

A compound that enters a cell changes what that cell transcribes. Which genes move, by how much, and in which direction is the most direct readout available of what the compound actually does — not what it was designed to bind, but what the cell does in response. Reading that response is routine. Reading it for every combination that might matter is not. A compound behaves differently at different doses, and differently again depending on the genetic background of the cell receiving it, so the space to be explored is the product of three large sets rather than any one of them. Even the largest screens sample a small corner of it, and every combination left untested is a question left open.

This is the gap prediction is meant to close. If the transcriptional response to a treatment could be computed rather than measured, the untested corners of that space would become inspectable, and the experiments actually run could be chosen because they are informative rather than because they happened to fit on the plate. The task, stated plainly, is this: given the transcriptome of an untreated cell and the identity of a drug, predict the transcriptome that cell would have shown had the drug been applied.

The task is harder than it sounds, for a reason that is easy to state and impossible to engineer around. The ground truth is a counterfactual. Sequencing destroys the cell it reads, so the treated and untreated profiles of the same cell can never both be observed, and a prediction can only ever be checked against a different cell that received the drug. Everything downstream inherits this. It is why paired training data must be assembled by matching cells rather than by tracking them, why evaluation is usually framed at the level of populations rather than individuals, and why distinguishing a real drug effect from ordinary cell-to-cell variation requires care rather than optimism.

Assembling data at the scale this problem demands became possible only recently. Two decades of falling sequencing costs, rising throughput and increasingly clever sample multiplexing turned single-cell profiling from a delicate experiment into something that can be run industrially, and Tahoe-100M [1] is what the end of that trajectory looks like: roughly a hundred million single-cell transcriptomes spanning some 1,100 small molecules across 50 cancer cell lines. Every compound and every cell line appears not as a summary statistic but as a cloud of individual cells.

That resolution changes which questions can be asked. The obvious one is conditional — what does this compound do to this cellular context, rather than what does it do on average across contexts that may have little in common. The less obvious question is whether the first has an answer. Because many conditions were run more than once, it becomes possible to measure how much of an apparent drug effect reproduces across replicates and how much is simply the variability of the assay. That second measurement turns out to matter as much as the first, and a good deal of what follows rests on the observation that standard evaluations quietly confuse the two.

The other development behind this thesis has nothing to do with cells. Over the same period, pretrained transformer language models became unreasonably good and unreasonably general, and the engineering poured into them — architectures, training recipes, scaling laws, the entire surrounding toolchain — came to exceed what any single scientific field could plausibly assemble for itself. That created a standing temptation: if a problem can be made to look like text, it inherits all of that work for free.

Cell2Sentence [2] takes the temptation about as literally as it can be taken. Rank the genes in a cell from most to least expressed, write their names out in that order, and the cell becomes a sentence — ordinary text that an ordinary language model can be fine-tuned on. The encoding is unsentimental about biology, discarding how much each gene is expressed and keeping only the ordering, on the argument that rank retains enough of the signal for the rest to be reconstructed. In exchange it requires no new architecture and no new training machinery, only cells written in a format that existing models already consume.

C2S-Scale [3] develops that idea into a family of models. Larger backbones are pre-trained on over 50 million cells drawn from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas, and then fine-tuned across a spread of tasks: predicting cell type and tissue, generating cells conditioned on a label, captioning clusters, answering questions about a dataset in natural language, and predicting perturbation responses. That last task is the one this thesis builds on, and it is worth being specific about its scope. Their perturbation work covers cytokine stimulation in peripheral blood cells, and, for small molecules, bulk L1000 profiles and the SciPlex3 panel. The single-cell results are scored as distributions — how closely the generated population for a condition resembles the observed one — rather than condition by condition. The checkpoint used throughout this thesis, **C2S-Scale-Pythia-1b-pt**, is the pretrained model from that family: atlas data only, with no drug-treated cells among it.

What we do is extend that line of work to chemical perturbation at the scale Tahoe-100M now makes available, and evaluate it as a prediction problem in the strict sense — condition by condition, against controls chosen so that a model which ignores the drug cannot pass. The question is whether a cell-sentence model fine-tuned on this atlas learns what a particular compound does to a particular cell.

Two things are worth keeping straight about what the model has actually seen, since the scale of the atlas invites an easy overstatement. Its pretraining is large but drug-free: tens of millions of cells, none of them treated with anything. And the fine-tuning does not consume Tahoe entire — it draws a sample of a few hundred thousand treated cells from it, which is ample for the task but is not a hundred million. Whatever the model learns about drugs, it learns during fine-tuning, from that sample.

It does learn something, and the something is the difficulty. The model acquires a good account of what a treated cell looks like *in general* — the shared shape that

perturbed transcriptomes have in common, which is substantial and largely the same whichever compound produced it. What it does not acquire is the part that distinguishes one compound from another. Change the drug named in the prompt while holding the input cell fixed, and the prediction barely moves; on the calibrated metric the model is statistically indistinguishable from a baseline that copies the untreated cell and is told nothing about any drug at all.

A second comparison is worth separating from that one, because it is weaker but more informative. A per-drug average, estimated from the same compound in *other* cell lines, also outperforms the model — but this is not a naive baseline. It is very nearly the best any method can do at the part of the problem it addresses, scoring 0.963 against a replicate ceiling of 0.968, and being close to it would be a real achievement rather than a low bar. What it cannot do is condition on the cell in front of it, which is precisely why the comparison is useful: it separates the drug’s average effect from the part that depends on context. Section 4.9 measures that split, and finds the second part worth roughly half the drug-specific signal — reached by the average, by the model, and by the literature alike, which is to say not at all.

A negative result earns its place only if it can be made specific, and that is the organising principle of what follows. We establish first that the failure is real rather than an artifact of how this field measures success — a distinction less pedantic than it sounds, since the standard metric can be saturated by a predictor that knows nothing about drugs. We then locate where in the pipeline the failure originates, show that the location implies a repair, apply it, and finally bound how much of the problem any method of this kind could recover even if that repair were perfect. The thesis therefore ends not in closure but in a number: a measured fraction of the drug-specific signal claimed by nothing in the literature, by no baseline we could construct, and by none of our own models.

Four questions carry that argument, in order:

1. **Is the evaluation trustworthy?** Before asking whether a model is any good, one has to ask whether the metric could tell. Can a predictor carrying no drug information score well on the measures the field has standardised on, and if so, which measures survive?
2. **Does the model use the drug at all?** Holding the input cell fixed and changing only the compound it is told about, does the prediction move — and does it move in the direction the real response moved?
3. **If not, where does the failure live?** Is the drug missing from the data, missing from the model’s internal representation, or present in that representation but ignored by the process that writes the output? Each answer implies a different repair, and only one of them is cheap.

4. **How much is recoverable at all?** A per-drug average is a strong baseline. How much of the drug-specific signal does it capture, how much is genuinely conditional on cellular context, and how much does any method — ours or anyone else’s — actually reach?

The contributions follow the same order. We audit the calibration of the standard evaluation and show that it can be saturated by a predictor carrying no drug information whatsoever, which bears on any work built from the same family of metrics. We establish drug-blindness on the one metric that survives that audit, using several independent instruments so that no single measurement carries the claim alone. We identify the target encoding — not the data, the architecture, or the training objective — as the binding constraint, and derive from that diagnosis a re-encoding of the prediction target which produces the first measurable drug effect in this project and generalises to drug-and-cell-line pairings never seen during training. And we separate the per-drug main effect from the drug-by-cell-line interaction, showing that the interaction accounts for roughly half the drug-specific signal and is reached by neither the lookup baseline nor the model.

The thesis proceeds as follows. Section 2 places the work against single-cell foundation models, the perturbation-prediction literature, and an unsettled dispute within it about whether deep models beat trivial baselines at all — a dispute this thesis adjudicates in one corner, using the instruments of the side it ultimately argues against. Section 3 covers the data, the model, the calibration framework, and, at some length because they carry the argument, the controls that make each negative result interpretable. Section 4 reports the findings in the order the argument needs rather than the order in which they arrived. Section 5 sets out the scope honestly, separating what is closed by measurement from what is merely untested, and Section 6 concludes.

2 Literature review

2.1 From text foundation models to single-cell foundation models

The models this thesis builds on descend from a template established in natural language processing. A transformer is pretrained on a very large unlabelled text corpus with a single objective — predict the next token — and the resulting network is then adapted to specific tasks by fine-tuning or prompting. The GPT line of models made the case that this recipe scales: capability on tasks never explicitly trained for improves steadily with model size, data and compute, so a single pretrained backbone can serve as a general substrate rather than a task-specific tool. What made the recipe portable is that it assumes almost nothing about language. It requires only that the data be expressible as sequences of discrete tokens, and that some tokens be predictable from others.

That portability is what invited the transfer into biology. If a cell can be written as a sequence, the same machinery applies unchanged. scGPT [4] and Geneformer [5] both take this route, treating a cell’s expression profile as a sequence of gene tokens and pretraining on tens of millions of cells before transferring to classification, integration and perturbation tasks. The ambition is usually stated as some version of the *virtual cell* [6]: a model whose simulated responses to intervention are trustworthy enough to substitute for an experiment.

The interesting variation among these models is not architectural but representational. A cell is a vector of expression values over some twenty thousand genes; a transformer consumes a sequence of discrete tokens. Something must be discretised or discarded in between, and what gets discarded bounds what the model can learn. The approach this thesis uses makes that choice unusually explicit.

2.2 Cell sentences, and the model this thesis fine-tunes

Cell2Sentence [2] takes the most literal route available: rank a cell’s genes by expression and write their symbols out in descending order. The cell becomes a *cell sentence*, which is text in the ordinary sense — a string that a tokeniser will happily consume:

MALAT1 ACTB TMSB4X B2M EEF1A1 RPL13A FTL TPT1 RPS27 GAPDH ...

Nothing marks this as biological. The information is carried entirely by which symbols appear and in what order, and a language model fine-tuned on such strings is doing exactly what it does on prose: predicting the next token given the previous ones.

The encoding keeps rank and discards magnitude. C2S defends the loss with a reconstruction argument: expression can be recovered from rank through a fitted log-linear relationship $e_i = a_d \log(r_i) + b_d$, which achieves $R^2 = 0.815$ against the original values

on peripheral blood data. On that basis they argue rank encodes expression reversibly enough across datasets for the discarded magnitude not to matter for downstream use. It is worth noting what kind of criterion this is — reconstruction fidelity is a statement about *absolute* accuracy, how close the recovered profile is to the true one, which is not automatically the same as preserving whatever distinguishes one experimental condition from another.

Understanding what the model does with such a sentence requires a little of the underlying architecture, because several of its properties matter later. C2S-Scale builds on decoder-only transformers of the GPT family, and the checkpoint used here is built on Pythia [7], a suite of such models released with open weights and training data. A decoder-only transformer factorises the probability of a sequence into a product of next-token predictions,

$$p(x_1, \dots, x_T) = \prod_{t=1}^T p(x_t \mid x_{<t}), \quad (1)$$

and is trained by minimising the cross-entropy between each predicted distribution and the token that actually occurred. Each token is first mapped to a vector by an embedding matrix, and those vectors form what is usually called the *residual stream*: a running representation, one vector per position, which every layer reads from and writes back into additively. A layer comprises multi-head self-attention, which lets each position gather information from earlier positions and is *causally masked* so that position t attends only to positions $\leq t$, followed by a position-wise feed-forward network. Both are wrapped in normalisation and added back to the stream rather than replacing it. After the final layer an unembedding matrix projects the stream to one score per vocabulary entry, and a softmax turns those scores into the next-token distribution.

Three properties of that design are worth noting now, since later sections depend on them. The residual stream is additive and shared, so a representation can be read at a chosen depth, modified, and the remaining computation allowed to proceed on the modified value — which is what makes interpretability interventions of the kind used in Section 4.3 possible at all. The training loss is a sum of per-token cross-entropies that weights every position equally, so a target whose tokens are largely shared between conditions gives any individual condition only a small share of the gradient. And conditioning is mediated entirely by attention across a causal mask, so information supplied early in a prompt must be reached from every later position at which it is needed.

C2S-Scale [3] develops Cell2Sentence into a family of models on this architecture, spanning backbones from Pythia at 410 million and 1 billion parameters up to Gemma-2 at 27 billion. Pretraining uses over 50 million human and mouse cells drawn from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas. The pretrained models are then fine-tuned across a range of tasks: predicting cell type and tissue, generating cells conditioned on a label, captioning clusters, answering natural-language questions about

a dataset, and predicting perturbation responses.

The perturbation task is the one this thesis extends, and its published scope is worth stating precisely. C2S-Scale evaluates it on three benchmarks: cytokine stimulation of peripheral blood cells, bulk L1000 profiles, and the SciPlex3 small-molecule panel. In each the model receives an untreated profile together with a description of the treatment and produces the treated profile, so the setup is conditional rather than label-only generation. Two features of it are relevant to the comparison this thesis draws. Pairing is by group rather than by cell — for the bulk benchmark, treated and untreated samples are matched by cell line name — which is unavoidable given that sequencing destroys the cell it reads. And the single-cell results are scored as distributions: for each condition the model generates as many cells as were observed, and the generated population is compared against the observed one, which asks whether the model produces a plausible population for a condition rather than whether it distinguishes that condition from another.

The checkpoint fine-tuned in every experiment below is **C2S-Scale-Pythia-1b-pt**, the pretrained member of that family, whose training corpus is atlas data alone and contains no drug-treated cells.

2.3 The dispute: do these models beat uninformative baselines?

The field contains an unresolved and unusually sharp disagreement about whether any of this works, and this thesis is best read as an attempt to settle one corner of it.

On one side sits a series of benchmarks reporting that deep perturbation models do not beat trivial predictors. Ahlmann-Eltze, Huber and Anders [8] find that deep-learning prediction of genetic perturbation effects does not outperform simple linear baselines. Csendes et al. [9] reach the same conclusion for foundation cell models on post-perturbation RNA-seq, and Bendidi et al. [10] put it most bluntly in a title, reporting that a single PCA still rules them all. Kernfeld et al. [11] concur in a broad comparison of expression-forecasting methods. The pattern is not confined to transcriptomic readouts: on scalar drug sensitivity, Hauptmann and Kramer [12] hold the experimental protocol fixed across architectures and find that the measured differences between them largely dissolve, and DrEval [13] finds that deep models barely outperform a naive predictor built from mean drug and cell-line effects. The recurring result is that a per-perturbation mean is remarkably hard to beat.

Against this, Miller et al. [14] argue that these conclusions “largely stem from limitations of benchmarking metrics, not from the models themselves”, and the argument is methodological rather than rhetorical. They formalise the use of positive and negative controls in perturbation benchmarking and introduce a new positive control, the *interpolated duplicate*, which blends a technical duplicate with the mean baseline using weights drawn from per-gene differential-expression significance — giving a reference point of known and

tunable quality. They then define a quantitative measure of whether a metric can see that quality at all, the *Dynamic Range Fraction* (DRF), and apply it across 14 Perturb-seq datasets and 13 metrics. Widely used measures such as MSE and control-referenced Pearson correlation turn out to be poorly calibrated, while weighted and rank-based alternatives — weighted MSE, weighted $R^2\Delta$ [15], and Normalised Inverse Rank [16] — are consistently calibrated. Evaluated on the metrics that survive, they report, deep models *do* beat mean, control and linear baselines.

What is actually in dispute is narrower than it first appears, because both sides agree the standard metrics are defective. The disagreement is about what a corrected evaluation would reveal. One reading is that the models were always capturing real biology and the broken metrics were hiding it, so better measurement should expose genuine skill. The other is that the models were never capturing much, and better measurement will simply show that failure more clearly and more honestly. Put that way the question is empirical rather than interpretive — but it is settled only within the setting where it is tested, and the setting of this thesis has been tested by neither camp. Miller et al. and Ahlmann-Eltze et al. both study **genetic** perturbation, where an intervention is a designed, near-binary knockout with a named target gene. DrEval studies **drug sensitivity** as a scalar potency such as IC_{50} , predicted from cell-line omics. None of them studies a full transcriptomic response to a small molecule — which acts at a dose and typically engages several proteins at once — at single-cell resolution, produced by an autoregressive generative model.

The position taken here is to adopt the proponents’ instrument rather than argue with it. Section 4.1 ports the DRF framework onto drug perturbation and uses it to decide which metrics are admissible in this setting before any model is scored, so that the evaluation is chosen by the sceptics’ opponents rather than by us. Whatever the models then turn out to do, the measurement cannot be dismissed by the objection Miller et al. themselves raise.

2.4 Six ways a drug-response benchmark can mislead

DrEval [13] deserves separate treatment, because beyond its verdict it supplies a diagnostic vocabulary that this thesis adopts throughout. Auditing drug response prediction, it identifies six obstacles to clinical translation: non-reproducible models, data leakage producing weak generalisation, pseudoreplication, biased evaluation metrics, missing ablation studies, and inconsistent response data. Four of them shape the methodology of Section 3 directly.

Pseudoreplication, in Hurlbert’s original sense [17], is treating repeated measurements of one unit as independent observations. DrEval observes that GDSC2’s 395,025 measurements descend from only 969 cell lines and 287 drugs, so an interval computed over measurements is badly overconfident about how much independent evidence exists. Our

data has the same shape — hundreds of thousands of cells resolving to a few thousand conditions across 50 cell lines — and every confidence interval in this thesis is consequently a clustered bootstrap over cell lines rather than over cells.

Biased evaluation they frame as a version of Simpson’s paradox. Because different drugs act at vastly different concentrations, most of the variance in a response dataset lies *between* drugs, so a model that learns nothing but each drug’s mean will explain most of it. Their illustration is stark: a global Pearson correlation of 0.90 alongside an average within-drug Pearson of 0.01. Section 3.3 explains why a discrimination metric evaluated *within* a comparison set is immune to this by construction.

Their treatment of splits is the part we borrow most directly. DrEval names four, each corresponding to a different application: Leave-Pairs-Out for missing-value imputation, Leave-Cell-Lines-Out for personalised medicine, Leave-Drugs-Out for drug design, and Leave-Tissue-Out for repurposing. Our holdout maps onto two of these — `unseen_combo` is Leave-Pairs-Out and `unseen_drug` is Leave-Drugs-Out — and Section 3.8 uses their names. The mapping carries a warning we take seriously. Citing Partin et al. [18], DrEval reports that good Leave-Pairs-Out and Leave-Cell-Lines-Out performance is reachable simply by predicting each drug’s average response across the training cell lines, so a strong result on those splits demands that the per-drug average be measured explicitly rather than assumed away.

Their remedy for missing ablations is randomisation and robustness testing. The scramble ablation of Section 3.4 is such a test, specialised to the drug: hold every byte of the prompt fixed except the compound, and see whether anything downstream notices.

2.5 Connectivity mapping and signature transfer

The Connectivity Map programme [19, 20] built an entire drug-discovery methodology on a specific premise: that a compound induces a characteristic transcriptional signature stable enough across cellular contexts that querying a reference database by signature similarity is informative. The L1000 platform [20] operationalised this at scale, and its 978-gene landmark set defines the gene panel used here, intersected with Tahoe’s measured genes to give 946.

The premise — that a drug’s signature transfers across cell lines — is normally assumed rather than measured, and how much of it holds determines whether a conditional model has anything left to predict once a per-drug constant is accounted for. Section 4.9 measures it directly, corrected for measurement noise, at single-cell resolution.

2.6 Decodable, used, and the global workspace

Linear probing establishes that information is *present* in a representation, not that it is *used* by the computation, and the distinction is central to the analysis in Section 4.3.

Elazar et al. [21] make the methodological case for keeping the two apart: probe accuracy licenses no conclusion about behaviour, and the way to ask whether a property is used is to remove it from the representation and measure what the model then does. Their amnesic-probing design is the one adopted here, with the addition of a matched control, since an intervention that fails to move the output proves nothing unless a comparable intervention does.

A recent line of work gives that distinction a sharper form. Gurnee et al. [22] report that language models maintain a small privileged set of representations — which they term the *J-space*, identified by a Jacobian Lens that measures the average linearised effect of an activation on the model’s likelihood of producing a particular token — and argue it functions as a global workspace analogous to conscious access. The J-space is strikingly small: roughly ten to twenty-five concepts are active at any moment, typically accounting for under 10% of activation variance. Its defining property is causal rather than correlational, and they establish it by intervention: swapping one active workspace vector for another changes the model’s answer to match. Content in the J-space drives behaviour; content outside it may be richly represented and still inert with respect to what the model actually emits.

That framing matters here because it supplies both a hypothesis and an experimental design. The hypothesis is that a property can be decodable from the residual stream at high accuracy while sitting outside the workspace the generation readout consults, in which case no amount of encoding it more explicitly will change the output. The design is the swap itself: overwrite one candidate representation with another and ask whether the model’s behaviour follows. Section 4.3 constructs a domain analogue of exactly this test, substituting a drug’s learned representation for another drug’s and measuring whether the emitted profile moves toward the substituted drug.

The related account of why a model might allocate its representations this way comes from Elhage et al. [23], who show that networks reserve dedicated directions for features the computation needs on every forward pass and push the remainder into superposition, where a feature remains linearly recoverable without occupying an axis of its own. Read together, the two suggest an asymmetry worth testing: a property required to emit any plausible output should sit in the causally active set, while one needed only for occasional, context-dependent composition need not.

2.7 Optimal transport for state matching

Predicting a treated cell from an untreated one runs into an obstacle that is not statistical but physical: sequencing destroys the cell it measures. No cell is ever observed both before and after treatment, so there is no ground-truth correspondence between the control population and the treated population — only two clouds of points in expression space,

sampled from the same well at different times. Any construction of paired training data must therefore *infer* a correspondence, and optimal transport is the standard framework for inferring one.

The problem in its classical form asks for the cheapest way to move one distribution of mass onto another. Monge’s original formulation seeks a map $T : \mathcal{X} \rightarrow \mathcal{Y}$ pushing a source measure μ onto a target ν , so that $T_{\#}\mu = \nu$, while minimising

$$\int_{\mathcal{X}} c(x, T(x)) \, \mathrm{d}\mu(x), \quad (2)$$

where c is a ground cost, typically squared Euclidean distance. This is intuitive — every source point is assigned one destination — but it is frequently ill-posed, because a deterministic map cannot split mass, and no such map exists when the target is more spread out than the source. Kantorovich’s relaxation resolves this by replacing the map with a *coupling*: a joint distribution $\pi \in \Pi(\mu, \nu)$ whose marginals are μ and ν , giving

$$\min_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) \, \mathrm{d}\pi(x, y). \quad (3)$$

A coupling may send fractions of a source point to several destinations, which is what makes the relaxed problem always solvable and appropriate for noisy, overlapping empirical distributions. When c is the p -th power of a metric the optimal value defines the p -Wasserstein distance $W_p(\mu, \nu)$; the case $p = 2$ is the one used here, and its geodesics give a principled way to interpolate between distributions.

Solved exactly, the discrete problem is a linear program with roughly $O(n^3 \log n)$ cost, which is prohibitive at single-cell scale. Entropic regularisation makes it tractable. Adding a negative entropy term to the objective,

$$\min_{\pi \in \Pi(\mu, \nu)} \langle C, \pi \rangle - \varepsilon H(\pi), \quad (4)$$

renders the problem strictly convex, so the optimum is unique and takes the scaling form $\pi = \text{diag}(u) K \text{diag}(v)$ with Gibbs kernel $K_{ij} = \exp(-c(x_i, y_j)/\varepsilon)$. The Sinkhorn–Knopp algorithm recovers u and v by alternating matrix–vector products [24], which are cheap and parallelise well on a GPU.

The regularisation parameter ε is not merely a computational convenience, and it is the reason this framework appears in this thesis at all. It controls how sharply mass is assigned. As $\varepsilon \rightarrow 0$ the coupling concentrates and approaches the unregularised solution, in which each control cell is matched to essentially one treated cell. As ε grows the plan blurs, and in the limit every control cell is coupled equally to every treated cell — the product of the marginals. Given a coupling, the *barycentric projection* maps each control

cell x_i to the π -weighted average of its partners,

$$\hat{y}(x_i) = \frac{\sum_j \pi_{ij} y_j}{\sum_j \pi_{ij}}, \quad (5)$$

and the two extremes of ε therefore produce two familiar objects: at small ε a specific observed treated cell, and at large ε the group mean. A sweep over ε is thus a continuous interpolation between two target constructions that would otherwise have to be treated as unrelated alternatives, which is how Section 4.4 uses it.

Applied to biology, this machinery has largely been used for trajectory inference. Waddington-OT [25] treats the cell population at each time point as a distribution in expression space and recovers a probabilistic ancestor–descendant link between snapshots, on the assumption that cells follow paths minimising transport cost. Subsequent work relaxes the assumption that mass is conserved — cells divide and die, so the two populations need not have equal total mass — using unbalanced formulations built on the Wasserstein–Fisher–Rao metric. A separate line of work exploits the fact that when both measures are Gaussian the transport cost has a closed form, the Bures–Wasserstein distance, which splits cleanly into a shift in means and a change in covariance structure and so distinguishes a population that moved from one that merely spread out.

Two practical points govern how couplings are computed here. The ground cost must be evaluated in a space where Euclidean distance is meaningful, and raw counts are not such a space: they are sparse, overdispersed and dominated by library size. Couplings are therefore computed in the released Tahoe scVI latent [26], a variational autoencoder embedding that models counts with a negative-binomial likelihood and yields a compact continuous representation in which distances behave. And because the cost of a full coupling grows quadratically, transport is solved within matched groups — a single cell line on a single plate — rather than across the atlas, which is both computationally necessary and biologically correct, since a control cell from one cell line is not a plausible counterfactual partner for a treated cell from another.

3 Methods

3.1 Data

Corpus. Tahoe-100M [1]: drug-treated single-cell transcriptomes across cancer cell lines, with plate-matched DMSO controls. Metadata supplies cell line (Cellosaurus ID), drug, dose (`drugname_drugconc`), plate, and a fine-grained mechanism-of-action annotation (`moa-fine`).

Gene panel. All analysis is restricted to the 946 genes in the intersection of the L1000 landmark set [20] with Tahoe’s measured genes. Expression is library-normalised to counts-per-10k and log $1p$ -transformed.

A representational pitfall worth recording. Tahoe’s per-cell `genes` field holds *vocabulary token identifiers*, not positional indices into the gene metadata table. Mapping them positionally scrambles gene identity silently — values exceed the metadata row count, so the error surfaces only as an out-of-bounds index at the tail. All expression construction here maps `gene_metadata['token_id']` $\rightarrow$ gene symbol $\rightarrow$ panel index. Validation: a PCA cell-line probe on the resulting panel reaches 0.864 against a chance rate of 0.02.

Unit of analysis. A *condition* is a (drug, cell line, plate, dose) tuple. Plates are numbered within cell line rather than globally, so the batch unit is the pair (cell line, plate) and never the plate label alone — a distinction that matters for every within-plate comparison below.

Scale. Table 1 states the quantities every n in this thesis should be located against. Two populations are used and they are not interchangeable. The broad characterisation of the task (Section 4.2) streams a wide sample of the atlas; the residual arm (Sections 4.5 onward) works from a smaller cached subset built once and reused, so that every arm sees identical cells.

Splits. Three evaluation tiers, defined by what the training set contained. *Tier 1* holds conditions whose (drug, cell line) pairing was seen in training, so it measures fit rather than generalisation. *Tier 2* holds conditions whose *drug* never appeared, which is Leave-Drugs-Out in the vocabulary of Section 2.4. *Tier 3* holds pairings whose drug and cell line were each seen but never together — Leave-Pairs-Out. Tier 3 is small (99 conditions) and is reported only where it is adequately powered; the residual arm builds its own, larger, (drug, cell line)-keyed holdout for the same purpose (Section 3.8).

Task characterisation (streamed sample)	
conditions characterised	
of which statistically active against DMSO	3,647 / 5
identifiable in principle (ceiling ≥ 0.8)	
with a plate-mate closer than their own replicate	
median cells per condition	
median #DEGs per condition (BH $q < 0.05$)	
median SNR (effect $\div$ replicate noise)	
drugs with a real dose series	
drugs seen in ≥ 3 cell lines	
Residual-frame cache	
cells	620,564 (600,000 treated)
(cell line, plate) groups	
drugs / cell lines	
conditions with ≥ 40 treated and ≥ 20 control cells	
surviving the reliability filter ($\cos(r_A, r_B) > 0.2$)	
training examples (60 control cells per condition)	
Evaluation sets	
tier 1 (seen conditions) / tier 2 (unseen drugs) / tier 3 (unseen combinations)	
conditions scored per tier in the generation evaluations	
tier-2 drugs in the expression-frame NIR benchmark (within plate)	
conditions scored in the residual-frame evaluation	570 (41)

Table 1: The populations behind every n quoted in this thesis. Sources: `drug_biology_atlas.json`, `eval_endcell_cpu.json`, `nir_sameplate.json`, `re_holdout2_stratified.json`, `target_divergence.json`. Note that the 72.7% active figure is over the 5,014 conditions with enough cells for the permutation test, not over all 6,628.

3.2 Model and representation

Cell sentences. A cell is written as its expressed panel genes in descending order of expression, terminated by `[END_CELL]`. The prompt supplies cell line, drug, dose, mechanism, and the untreated control cell’s own sentence; the response is the treated cell’s sentence.

Backbone. `C2S-Scale-Pythia-1b-pt` [3], a C2S-Scale checkpoint built on the Pythia-1b decoder [7], cold-started for every arm from the same base checkpoint with identical hyperparameters (1 epoch, learning rate 10^{-5} , gradient accumulation 16, maximum length 8192, bf16), so that in each experiment the training target or objective is the only variable.

What that checkpoint has already seen bears directly on how the held-out splits should be read. It is pretrained on over 50 million human and mouse cells drawn from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas — an *atlas* corpus, carrying no

chemical or genetic perturbation data, and the released model card lists cell and tissue prediction, conditioned generation and gene-set conversion among its capabilities but not perturbation prediction. The base model therefore reaches our fine-tuning having never seen a drug-treated cell, which is what allows the tier-2 and tier-3 splits to mean what they claim: a drug held out here is unseen by the whole pipeline, not merely by its final stage.

Sentinels. [END_CELL] and, for the residual arm, [DOWN] are registered as atomic special tokens. This matters: unregistered, [DOWN] splits into sub-words and the up/down boundary ceases to be a clean symbol.

3.3 Metrics

The standard metric. The field’s default is a correlation between predicted and true *change from control*, restricted to the genes that actually move. Writing x for the control profile, y for the treated truth and $\hat{y}$ for the prediction, let

$$S_K = \arg \text{top-} K_g |y_g - x_g| \quad (6)$$

be the K genes with the largest true change. Then

$$\text{DE-}\Delta r = \text{corr}((\hat{y} - x)|_{S_K}, (y - x)|_{S_K}). \quad (7)$$

We report $K = 50$ with Pearson correlation throughout; $K \in \{20, 100, 200\}$ and the Spearman variant were computed and move the numbers by less than 0.02 except at $K = 200$, where the top- K set starts to include genes that do not move.

Three details are load-bearing rather than incidental, because Section 4.1 turns on them.

First, S_K is selected using the *truth*, so every predictor is scored on the genes that condition actually moved. Second, and consequently, $\text{DE-}\Delta r$ rewards agreement with $y - x$ and a prediction that simply reverts toward the population centre correlates with $y - x$ by construction — which is the artifact the results chapter exhibits. Third, **partial-DE- Δr** is the same quantity with the control profile regressed out of both arguments before correlating, which removes that route and is therefore the honest version.

Absent genes. A cell sentence lists only expressed genes, so a predicted profile leaves some panel genes unranked. Two conventions were used throughout and reported side by side: **worst** assigns every absent gene the last rank P , and **francesca** assigns them a fixed middle rank $P/2$. The choice is immaterial — spike-in $\text{DE-}\Delta r$ reads 0.952 against 0.954 under the two, and no baseline or ceiling in this thesis moves by more than 0.01 —

so all figures below are quoted under **worst** and the convention is not mentioned again.

panel- τ . Kendall’s τ between predicted and true rank over the full 946-gene panel. Unlike DE- Δr it selects no gene subset and is invariant to the absent-gene convention by construction.

Normalised Inverse Rank. Introduced by Wu et al. in PerturBench [16] (there called the “transposed-rank” metric) and identified by Miller et al. [14] as one of the few consistently calibrated metrics across 14 Perturb-seq datasets. For a prediction $\hat{y}$ of condition i , with truths $\{y_j\}$ over the comparison set,

$$\text{NIR}(\hat{y}, i) = \frac{1}{|J|} \sum_{j \in J} \left[\mathbf{1}\{\sigma(\hat{y}, y_i) > \sigma(\hat{y}, y_j)\} + \frac{1}{2} \mathbf{1}\{\sigma(\hat{y}, y_i) = \sigma(\hat{y}, y_j)\} \right], \quad (8)$$

the fraction of *other* conditions whose truth the prediction resembles less than its own, with ties counted at one half. Chance is 0.50. Because the generic response is present in every condition’s truth in equal measure, it cancels in the comparison: NIR is a *discrimination* metric, and only the drug-specific difference decides it.

The tie term is not cosmetic. In residual space (Section 3.7) a predictor of “the average drug response” is exactly the zero vector, every similarity against it is zero, and a strict inequality scores it 0.000 rather than the 0.500 it deserves.

Which NIR variant, and why it must be stated. Our instruments compute NIR under two similarity functions and store both: **nir_expr**, Euclidean distance in expression space, and **nir_rank**, the same construction over rank vectors. **Every NIR figure in this thesis is nir_expr**, and the choice is load-bearing rather than incidental: under **nir_rank** the tier-2 aggregate places *every* arm below chance, including the noise ceiling itself (0.380, against the model’s 0.313 and a drug-agnostic linear map’s 0.285). A metric on which a real replicate of a condition fails to be retrieved above chance is not measuring what its name suggests, and it is not the metric the calibration analysis of Section 4.1 endorsed — the DRF result is computed on the expression-space variant. We report the calibrated one and name it here so the stored artifacts cannot be misread.

Metric calibration (DRF). We port the framework of Miller et al. [14], in which each *metric* is graded rather than each model, by asking how well it separates a positive from a negative control:

$$\text{DRF}(m) = \frac{m(\text{pos}) - m(\text{neg})}{m(\text{perfect}) - m(\text{neg})}, \quad (9)$$

with pos an interpolated-duplicate noise ceiling — their positive control, which interpolates a technical duplicate toward the mean baseline with per-gene weights given by

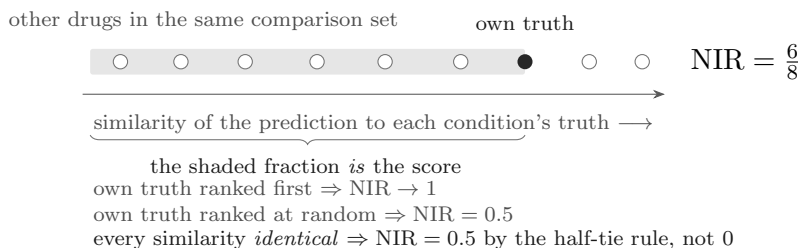


Figure 1: **NIR asks whether a prediction resembles its own condition’s truth more than it resembles the truths of the other drugs in the same comparison set.** The generic response — the stress and cell-cycle program every compound triggers — is present in every truth in equal measure, so it cancels in the comparison and only the drug-specific difference decides. Two consequences follow, and both matter downstream. Chance at 0.50 is a statement about the *comparison set*, not about the model: conditions are grouped by (cell line, plate) in the expression frame and by cell line in the residual frame, and the two are not interchangeable. And a predictor whose similarities are all identical — the zero vector, which is what “predict the average drug response” becomes in residual space — must score 0.50 rather than 0, which is why ties are counted at one half rather than dropped.

differential-expression significance — and neg a drug-agnostic leave-one-out mean. A negative DRF means the metric scores the uninformative baseline *above* the noise ceiling: it is inverted, and rewards the generic gene ordering rather than the drug.

Adopting this framework is deliberate. Miller et al. use it to argue that reports of deep models failing to beat uninformative baselines reflect miscalibrated evaluation rather than model failure (Section 2.3). Evaluating our model on the metric *their* analysis endorses means the result reported in Section 4.2 cannot be dismissed on the grounds their own paper raises.

3.4 Controls

The negative results in this thesis are only as strong as the controls that make them interpretable. Each of the following was added in response to a specific objection.

Plate confounding. Drug and plate are partially confounded by Tahoe’s design: each drug is assayed on its own plate(s), with roughly 4–12 drugs per plate and plate-matched controls. Discrimination grouped by cell line alone therefore lets the *batch* signature stand in for drug identity. Measured directly, by scoring the same tier-2 conditions with and without the plate held constant:

The mean baseline is the clearest demonstration: it contains no drug information whatsoever, and it scores 0.399 when the comparison set may span plates against 0.161 when it may not. Every discrimination instrument therefore gained a `--same_plate_only` mode and all discrimination results in this thesis were re-run within plate. Conclusions are unchanged; some magnitudes shrink.

Predictor (drug-agnostic)	cross-plate	within plate
mean baseline	0.399	0.161
control-copy	0.551	0.510
noise ceiling	0.625	0.575
fraction of drugs at chance	43%	49%

Table 2: Expression-frame NIR. A drug-agnostic mean baseline more than doubles its score when the plate is free to vary, which is what plate leakage looks like. The ceiling falls once the plate is held constant because the batch signature was inflating it too.

A separate and stronger observation, which belongs to the drug-blindness argument rather than to this one, is that *within* plate and on the identifiable stratum a control-copy baseline reaches 0.766 against the model’s 0.768 (Section 4.2). That is not a statement about plate leakage — it is a statement about the model.

The scramble ablation. The central manipulation. The prompt is left byte-identical except that the drug name and mechanism are replaced by another drug’s, while the control cell and the scoring truth are unchanged. Because both arms share the same control cell, control- and batch-derived leakage cancels in the *difference* $\Delta_{\text{scr}} = \text{NIR}_{\text{model}} - \text{NIR}_{\text{scramble}}$. This is the only manipulation that isolates drug use, and every drug-use claim in this thesis is anchored to it rather than to a comparison against a drug-agnostic predictor.

Two further drug-isolating instruments. The drug-blindness claim rests on four instruments rather than one, because they fail in different ways and agreement across them is worth more than any single number. Two are defined above (NIR and the scramble); the other two are:

Forced-choice grading. A prediction is scored against its own condition’s truth and against another drug’s truth from the same comparison set, and the instrument records how often the own truth wins. Chance is 0.50 by construction. The ceiling is the same forced choice with a real held-out replicate substituted for the prediction, which bounds what any predictor could achieve at this noise level; it lands between 0.67 and 0.83 depending on cells per condition.

Output invariance. This asks whether the drug token changes the generated text at all, without reference to any truth. For a fixed control cell, the model generates twice from the *same* prompt and once from each of several prompts differing only in the drug, and the instrument compares

$$\text{gap} = \underbrace{\overline{\text{sim}}(\text{same prompt, resampled})}_{\text{sampling noise alone}} - \underbrace{\overline{\text{sim}}(\text{prompts differing in the drug})}_{\text{noise plus any drug effect}}, \quad (10)$$

with similarity measured as top-100 Kendall τ and as Jaccard overlap. A positive gap means two generations for the same drug resemble each other more than generations for different drugs do, i.e. the drug token moves the output. A gap of zero means swapping the drug perturbs the output no more than resampling at the same temperature does. Reported over 8 contexts $\times$ 12 drugs at temperature 0.8. This instrument is the weakest of the four — it can only detect whether the output moves, not whether it moves *correctly* — but it is also the one least able to be confounded, since it needs no ground truth at all.

Swap similarity — stratified scramble. A scramble that swaps in a drug with a *similar* true response is a weak test: an unchanged output would then be correct, not blind. This is a real risk here, because same-mechanism drugs are barely more similar than random pairs (within-MoA distance 2.322 versus between-MoA 2.377, ratio 0.977), so “different mechanism” does not imply “different response”. Two remedies are used:

1. a *swap-distance sweep* that turns the single test into a curve of Δ_{scr} against the response distance $\|y_A - y_B\|$ between the real and swapped-in drug; and
2. a *stratified scramble* with three defined partners per condition chosen by residual cosine within the cell line — **near** (most similar), **orth** ($\cos \approx 0$) and **opposite** (most anti-correlated). Genuine drug use must produce a gap that *grows* across the strata; a flat profile is a red flag.

Field decomposition of the scramble. The scramble above replaces the drug name *and* its mechanism together, so a null tells us the instruction was not read without saying which part of it was ignored. Decomposing it costs one extra pair of generation arms and answers a question the whole unseen-drug argument turns on. Each arm swaps exactly one span of the instruction line to the opposite-signature partner and leaves every other byte identical: the drug name with the mechanism kept, the mechanism with the name kept, and — for continuity with earlier results — both.

One guard is essential rather than tidy. If the swap partner happens to share the original’s mechanism, then swapping only the mechanism produces a prompt *identical* to the model’s own, and the arm scores a gap of exactly zero *by construction*. Read naively that is indistinguishable from “the model ignores the mechanism”, which is the very claim under test. Such conditions are therefore dropped rather than scored, which is why the mechanism arm carries fewer conditions than the others.

Channel gate. For an unseen drug, prediction is only possible through a property the drug *shares* with drugs seen in training, so the reachable channels can be enumerated and each one measured in the same frame. For a channel X , a drug’s residual is predicted as the mean residual of the *other* drugs in the same cell line that share property X with it

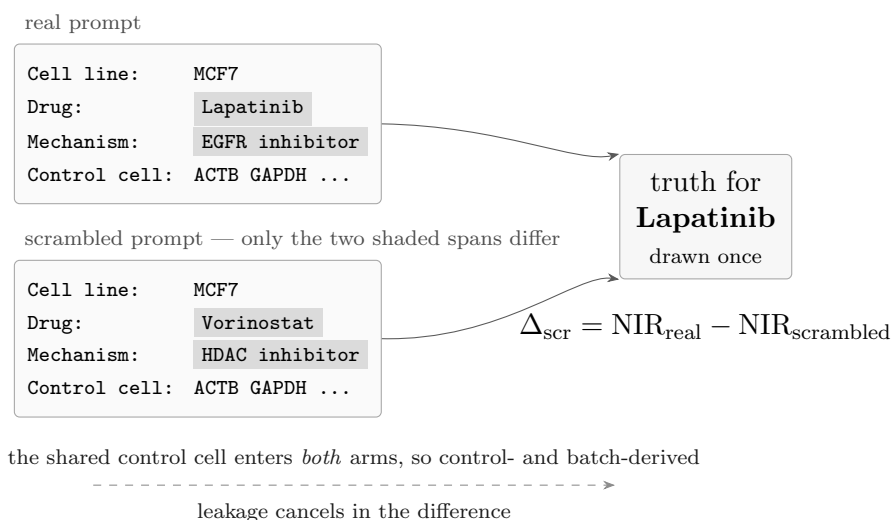


Figure 2: **The scramble replaces the drug name and its mechanism and nothing else.** Both arms are conditioned on the same control cell and scored against the same truth, so any leakage through the control cell or the plate enters both arms in equal measure and cancels in the difference Δ_{scr} . This is why Δ_{scr} , rather than a comparison against a drug-agnostic predictor, is the quantity every drug-use claim in this thesis is anchored to: the drug-agnostic comparison is not leak-immune, and a drug-agnostic mean baseline scores $\text{NIR} = 0.399$ when the comparison set may span plates against 0.161 when it may not, precisely because it inherits that leakage. A refinement matters here and is developed in Section 3.4: if the swapped-in drug happens to have a *similar* true response then an unchanged output is correct rather than blind, so the swap partner is chosen from three defined similarity strata and the evidence is the *gradient* across them rather than any single value.

— the construction of the mechanism baseline, generalised. Because it uses only drugs that were in training, it is a fair contest in every split, unlike a same-drug lookup, which on an unseen drug is an oracle.

Two disciplines make the result readable. Every channel is paired with a **count-matched random null**: the same *number* of partner drugs, drawn at random from the same cell line and averaged identically. Without it, a channel scoring above chance cannot be distinguished from the fact that averaging k residuals denoises, which is true for any k and says nothing about the channel. And **coverage is reported before any verdict**, because a channel annotated on few drugs cannot be *closed* by a null result — it was never tested.

The bound this establishes is on *similarity transfer* through each channel, not on any model. A model could in principle combine channels or learn a non-smooth structure-to-response map that a nearest-neighbour average cannot express. A leave-one-drug-out regression over all channels jointly addresses the first. The second is bounded by sample size rather than by argument: at a few hundred drugs, learning a non-smooth map is not feasible, so at this n the lookup is effectively the bound.

Is the interaction learnable? The transfer coefficient says how *large* the drug×cell-line interaction is; it says nothing about whether anything could learn it. If a cell line modulated drug responses in a consistent direction, that direction would be estimable from the line’s own cells and a conditional model would have a concrete target. Writing $\kappa(d, c) = r(d, c) - \hat{\beta}(d)$, the test compares $\cos(\kappa(d, c), \kappa(d', c))$ for different drugs in the *same* cell line against the same quantity across *different* cell lines.

$\hat{\beta}$ must be estimated **leave-one-cell-line-out**. Estimated over all lines it would contain $r(d, c)$ itself, and subtracting a mean containing its own term induces a $-1/(m - 1)$ correlation that biases the signal arm downward *by construction* — which would make a structured interaction look idiosyncratic, the exact error the test exists to avoid. Both arms are disattenuated by κ ’s own split-half reliability, since κ is a difference of two noisy quantities and is therefore noisier than the residual it comes from.

Uncertainty. All confidence intervals are clustered bootstraps resampling *cell lines*, not conditions. Conditions within a cell line are pseudoreplicates in Hurlbert’s sense [17] — multiple measurements of one unit treated as independent — and an observation-level interval would be badly anti-conservative. This is one of the six pathologies DrEval [13] identifies in this field, and the structure of our data is the same as the one they describe: hundreds of thousands of cells resolving to a few hundred conditions over 50 cell lines. For the transfer coefficient (Section 3.10) the cluster is the *drug*, since the estimand is a per-drug property.

Generation validity. Every generation-based evaluation reports the [END_CELL] / [DOWN] completion rate, the fraction of emitted symbols that are valid panel genes, and the duplicate rate, before any score is quoted. This is not hygiene: a 600-token generation cap silently truncated 26% of generations in an early run and *halved* the measured effect.

3.5 Mechanistic probing

To distinguish “cannot represent the drug” from “represents but does not use it”:

Decodability. Per-layer linear probes on the residual stream at the last prompt token, predicting drug identity.

Confound purification. The raw drug subspace is partly a batch subspace — probing dose and plate from *inside* it recovers dose at 0.90–0.95. A context subspace is built from the union of cell line, plate and dose, and the drug axis is orthogonalised against it. Every causal claim uses the purified subspace.

Removal gate. Before any “ablating the drug does nothing” result is trusted, the subspace must be shown to contain the drug: the fraction of above-chance probe accuracy removed by projecting it out must exceed 0.8. Without this gate the null is unfalsifiable.

Causal test. Log-probabilities of the teacher-forced true response are read clean, then re-read with a forward hook projecting the subspace out at every position; the statistic is the mean KL divergence over response tokens. Measuring in logit space rather than by generating removes the sampling-noise floor that made an earlier steering experiment inconclusive.

The identity test. Ablation shows a subspace carries functional variance; it cannot show that the subspace’s *drug-specific content* is read. Only a swap can. For each cell that truly received drug A, drug A’s code is overwritten with drug B’s *within the purified subspace*, and the resulting KL is compared against injecting a random vector of *identical norm*. Drug means are estimated on a held-out half of the cells and the swap performed on the disjoint half, so mean-estimation noise cannot itself manufacture a null.

3.6 Target constructions

Consensus. Cells grouped by (drug, cell line, dose) and averaged into one pseudobulk sentence, with every original prompt retained so only the response changes.

Optimal transport. For each (cell line, plate) group, a coupling π between control and treated cells minimising $\sum_{ij} \pi_{ij} \|x_i - y_j\|^2$ under uniform marginals, solved by Sinkhorn iteration [24] on $K = \exp(-C/\varepsilon)$. Couplings are computed in the released Tahoe scVI latent [26] ($n_{\text{latent}} = 10$), joined per cell by barcode rather than re-encoded — a re-encode produced a degraded latent (cell-line probe 0.11 versus 0.975 for the shipped latent). Targets are built *decoder-free*, as convex combinations of real treated cells in expression space, so they always lie on the data manifold.

The regularisation parameter defines a ladder whose endpoints are experiments we had already run: $\varepsilon \rightarrow \infty$ gives a uniform coupling and hence the treated mean (consensus); $\varepsilon \rightarrow 0$ gives near-hard assignment, approximately a single state-matched treated cell. This is what makes the sweep conclusive rather than anecdotal.

3.7 Residual targets

For each condition (drug, cell line, plate, dose) with at least 40 treated and 20 plate-matched control cells:

$$r(d, c) = \underbrace{(\bar{y}_{\text{treated}} - \bar{y}_{\text{control}})}_{s(d, c)} - \underbrace{\frac{1}{|D_c|} \sum_{d' \in D_c} s(d', c)}_{\text{generic}(c)}. \quad (11)$$

The control subtraction removes the cell’s own state; the mean-over-drugs subtraction removes the *generic drug program* — the stress and cell-cycle response common to all compounds, which Section 4.1 shows accounts for essentially all of the model’s apparent skill. What remains is what makes *this* drug different.

Scope, and a justification that did not survive. The generic is estimated per *cell line* rather than per plate. The original argument was reproducibility — 62% of conditions reproduce across a half-split at cell-line scope against 19% at plate scope — on the reasoning that a plate holds too few drugs to estimate a mean, so subtracting a noisy one injects noise.

That comparison does not hold up, and Section 4.9 retracts it. Both figures were computed with a *shared* control pseudobulk subtracted from both halves, which makes the halves correlate through their common control error — and it inflates cell-line reproducibility far more than plate reproducibility, because the cell-line generic is itself less noisy. Measured honestly, with disjoint control halves, the two are comparable: 20% at plate scope against 16% at cell-line scope. The reproducibility argument therefore favours neither, and three other diagnostics favour plate scope outright.

The targets used throughout this thesis were nonetheless built at cell-line scope, because that decision was made before the diagnostics existed. The consequence is stated

rather than hidden: the residual targets retain plate-level batch structure. This does not touch Δ_{scr} , which is a paired contrast in which any component shared by both arms cancels, but it does mean a plate-scoped rebuild is the obvious next construction.

Reliability filter. A condition is kept only if its residual reproduces across a half-split, $\cos(r_A, r_B) > 0.2$. This retains 4,091 of 6,617 conditions (62%); the remainder are noise.

Encoding. `<up genes> [DOWN] <down genes> [END_CELL]`, 100 genes per block, ordered by signed residual magnitude. Sign is biology, so it is given a symbol rather than being folded into a magnitude ranking.

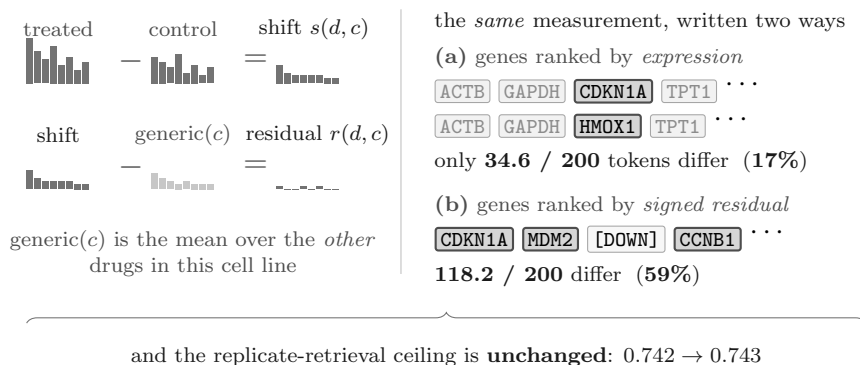


Figure 3: **The information was always present; the tokenisation discarded it.** Left: the drug-specific residual is built by two subtractions. The control removes the cell’s own state, leaving the *shift*; the mean over the other drugs in the same cell line removes the *generic* program that every compound triggers, leaving what makes this drug different. Right: that same measurement, written two ways. Ranking genes by expression (a) puts housekeeping genes first, so only 34.6 of the top 200 tokens differ between two drugs in the same context and cross-entropy has almost nothing to learn drug identity from; ranking by signed residual (b) raises that to 118.2. The bracket is the point of the figure — the replicate-retrieval ceiling is essentially identical under both encodings, so no information has been added. What changed is how much of it the token sequence exposes to the loss.

Scoring. Predicted and true residuals are both mapped to a signed rank vector (top- k up positive, top- k down negative, weight $1/\log_2(\text{rank} + 1)$) so a generated sentence and a continuous truth are compared like for like; similarity is cosine and the score is NIR over other drugs in the same cell line.

3.8 Generalisation design

Following DrEval’s insistence that the split match the intended application [13], and using its nomenclature, we build a three-way split keyed at the **(drug, cell line)** level so that every plate and dose of a held-out pair goes to the held-out split and no sibling condition leaks into training:

unseen_drug — **Leave-Drugs-Out (LDO)** every condition of a held-out drug. The model never sees the token. The application is drug design. This is the *control*: given that chemical structure does not predict response (Section 4.10), it should fail, and its failing is what proves that any transfer observed in **unseen_combo** comes from having seen the drug.

unseen_combo — **Leave-Pairs-Out (LPO)** a held-out (drug, cell line) pair whose drug is seen in at least three other cell lines. The application is missing-value imputation. This is cross-context transfer and the scientifically meaningful test.

train everything else.

We do not build a Leave-Cell-Lines-Out (LCO) split here; Section 5 argues it is the natural next axis and explains why its interest is contingent on the transfer coefficient.

A warning we inherit with the vocabulary. DrEval reports, citing [18], that apparently good LPO and LCO performance can be reached simply by predicting each drug’s average response across the training cell lines. Our transfer result is an LPO result and therefore sits exactly in that regime. This is the reason the baseline ladder below is not optional: the drug-average predictor is not a formality to be dismissed, it is the hypothesis under test.

Stratified sampling. Evaluation samples conditions under an explicit per-split quota. A uniform draw reproduces the pool’s proportions, which starves precisely the split the experiment exists to test: in a first run, 250 constructed held-out combinations yielded only 33 scored conditions, purely as an artifact of the sampler.

3.9 Baselines

Each is a predictor of the drug-specific residual, scored through an identical pipeline — same truths, same signed-rank encoding, same comparison set.

drug_lookup_1 selects a cell *line* and averages that line’s conditions rather than selecting a single condition, because 62% of drugs are multi-dose and a single condition would confound “different cell line” with “different dose”. Baselines are reported per split: on **unseen_drug** the same-drug lookups are *oracles*, since they read the held-out drug’s own residuals, information the model is definitionally denied. **moa_lookup** is not an oracle in any split, as it uses only drugs that were in training.

Baseline	Information used	Role
<code>drug_lookup</code>	this drug’s mean residual, all other cell lines	the bar to beat
<code>drug_lookup_1</code>	this drug in <i>one</i> other cell line	removes the averaging advantage
<code>moa_lookup</code>	other drugs sharing this drug’s MoA, same cell line	mechanism-leak diagnostic
<code>control_copy</code>	none (predict the control cell)	must be ≈ 0.50
<code>generic</code>	none (predict the average response)	must be ≈ 0.50
<code>random</code>	none	floor

Table 3: The baseline ladder. `drug_lookup` is the pre-registered win condition: a lookup can reproduce a memorised average, so only a model that tailors the response to the control cell’s state can beat it.

3.10 Variance decomposition

To separate the per-drug main effect from the drug×cell-line interaction we model

$$r(d, c) = \beta(d) + \kappa(d, c) + \epsilon, \quad \mathcal{T} = \frac{\text{var } \beta}{\text{var } \beta + \text{var } \kappa}, \quad (12)$$

where β is constant across cell lines and κ is the interaction — the only component a conditional model can win on.

This is DrEval’s naive predictor, in transcriptomic space. DrEval’s `NaiveMeanEffectsPredictor` predicts $\hat{y}_{ij} = \mu + (\mu_i^c - \mu) + (\mu_j^d - \mu) = \mu_i^c + \mu_j^d - \mu$: an additive model of cell-line and drug main effects, with no interaction term. Our `generic(c) + beta(d)` is precisely that quantity, lifted from a scalar potency to a 946-dimensional profile. It follows that κ is the term their naive model omits, and that $1 - \mathcal{T}$ measures exactly the headroom available above it. DrEval finds that deep models barely beat this additive baseline on drug sensitivity; the transfer coefficient asks whether, in the transcriptomic setting, there is anything above it to win in the first place.

$\mathcal{T}$ is estimated by disattenuated cross-line correlation,

$$\hat{\mathcal{T}} = \frac{\cos(r(d, c_1), r(d, c_2))}{\sqrt{\rho(d, c_1) \rho(d, c_2)}}, \quad \rho = \frac{2 \cos(r_A, r_B)}{1 + \cos(r_A, r_B)}, \quad (13)$$

with ρ the Spearman–Brown reliability of the full-sample residual. Dividing out the reliability is essential: measurement noise alone depresses the raw cross-line cosine and would otherwise be read as interaction.

Five controls accompany the estimate, each capable of changing the conclusion:

1. **Control half-split.** The production target builder subtracts one control pseudobulk from both halves, so control-estimation error is common to r_A and r_B ; this inflates the reliability and biases $\mathcal{T}$ *downward*, manufacturing apparent interaction. Both builds are run and the bias reported.

2. **Negative control, structure-matched.** A different drug in a *different* cell line, breaking only the drug match while preserving the cross-line structure of the estimand. Must be ≈ 0 . A same-plate/different-drug control was tried first and is *not* adequate: it differs from the estimand in two ways at once, and plate structure surviving a cell-line-scoped generic inflates it to $+0.478$ while leaving the estimand untouched — which produced a false-alarm rejection before the matched control was built.
3. **Simulated null.** At $\kappa = 0$ the estimator does not return 1.0. A simulation calibrated to the measured noise level fixes the reference. In validation the raw cross-line cosine at $\kappa = 0$ is 0.734, which read naively reports 27% interaction where the truth is zero.
4. **Dose strata.** Same drug, same cell line, different dose is not cross-line transfer; it is reported separately and serves as a reference scale.
5. **Leave-one-out generic.** The generic includes drug d , leaving a $-1/m$ self-term.

Validation against planted ground truth recovers $\mathcal{T} = 1.00$, 0.70 and 0.40 to within 0.005, and the different-drug negative control returns -0.001 .

Finally, $\|r\|/\|s\|$ reports what fraction of the total perturbation response the drug-specific frame covers — the number that scopes every claim in this thesis, and one that has not previously been computed for this task.

4 Results and Analysis

4.1 The metric is the first result

A zero-information baseline scores above the noise ceiling. A predictor that places every gene at the middle rank — carrying no information about drug, cell, or biology — scores $\text{DE-}\Delta r(K=50) = 0.9999$ [0.99991, 0.99992], above the two-real-cell noise ceiling (0.76) and above the model (0.73). The ordering is inverted: less information yields a higher score, which is the signature of an artifact rather than of skill. Note also that this predictor’s partial-DE- Δr and panel- τ are *undefined* rather than merely poor — its shift from control is identically zero, so there is nothing left to correlate once the control is regressed out. The cause is control regression-to-the-mean: DE- Δr correlates predicted and true change from control, and a prediction that reverts toward the centre correlates with the true change by construction.

Predictor	Information	DE- Δr (K50)	partial-DE- Δr
revert_center (all genes $\rightarrow P/2$)	none	0.9999	undefined
revert_mean (predict mean control)	none, no fit	0.961	0.25
ridge linear (control $\rightarrow$ shift)	control	0.947	0.28
noise ceiling (two real cells)	—	0.76	—
model (1B LLM)	control + drug	0.73	not measured

Table 4: DE- Δr is saturated by a control artifact. Regressing the control out of the two fitted baselines leaves ≈ 0.26 of real but *drug-agnostic* skill, and a baseline that performs no fitting at all reaches the same figure. The model’s own partial-DE- Δr was never computed — it requires a GPU pass that was not run — so the cell is marked rather than filled with the expected value; the model’s panel- τ , which *is* measured, carries the same conclusion below.

Removing the control (partial-DE- Δr) collapses the raw 0.95 to ≈ 0.26 for the two baselines that were scored — and that residue is matched by a baseline that performs no fitting at all. The model’s own partial-DE- Δr is not measured, so the comparison is anchored instead to panel- τ , which is measured for every arm and tells the same story: **revert_mean** 0.268, ridge 0.265, model 0.257 (tier 2, **worst** convention). The response decomposes into control reversion (an artifact), a generic drug program (≈ 0.26 , drug-agnostic, trivially matched), and a drug-specific component that no predictor captures.

Only one metric survives calibration. Under DRF computed within plate (61 groups, 655 drugs, ≈ 53 cells per drug):

The inversion is a property of the data, not a leak: a leave-one-out correction barely moves the numbers, since real cell lines carry 20–204 drugs. It also survives holding the plate constant. The one earlier claim that did not survive is retracted here: **weighted_r2** is approximately neutral rather than strongly inverted.

Metric	DRF (within-plate)	$m(\text{neg})$	$m(\text{pos})$	Verdict
NIR	+0.446	0.274	0.597	only calibrated metric
<code>weighted_r2</code>	-0.081	0.793	0.776	$\approx$ neutral
<code>panel-τ</code>	-0.231	0.694	0.623	inverted
<code>spearman_expr</code>	-0.415	0.851	0.790	inverted
<code>DE-Δr</code>	-0.448	0.886	0.834	inverted

Table 5: A negative DRF means a drug-agnostic baseline outscores the noise ceiling.

This replicates Miller et al. — and then diverges from them. Their central methodological claim is confirmed emphatically: metric choice dominates, the rank-based metric is the one that survives, and the correlation-based metrics they identify as poorly calibrated are exactly the ones that fail here. The agreement extends to the specific metrics — they find NIR consistently calibrated across 14 Perturb-seq datasets, and we find it is the only calibrated metric of the five we tested on drug perturbation. Our `weighted_r2` result is also consistent with their finding that weighted metrics behave better than unweighted ones.

The divergence comes at the next step. Miller et al. conclude that under calibrated metrics deep models outperform uninformative baselines; Section 4.2 evaluates on the calibrated metric and finds the model at chance. Two differences in setting are candidates for the discrepancy — genetic versus drug perturbation, and a discriminative regression model versus an autoregressive generative one — and Section 4.7 supplies a third and more structural possibility.

The metric can discriminate; the model cannot. A spike-in forced choice between two real drug populations separates them essentially perfectly at pseudobulk (`panel- $\tau$` = 1.000, `DE- $\Delta r$` = 0.995, top- n τ = 0.987, plate held constant). The instrument is not the bottleneck.

4.2 The model is drug-blind

Having established which instrument can be trusted, we can finally ask the question the thesis exists to answer. The answer is uniform across four instruments that fail in different ways, which is the only reason to report all four.

Which groupings carry information at all. Before asking what the model knows, it is worth asking what there is to know. A ladder of mean-shift baselines — each predicting the average response of a progressively narrower group — answers that directly.

Cell line roughly triples the global mean’s score; mechanism does not move it at any tier, and MoA $\times$ cell line does not improve on cell line alone. So the informative

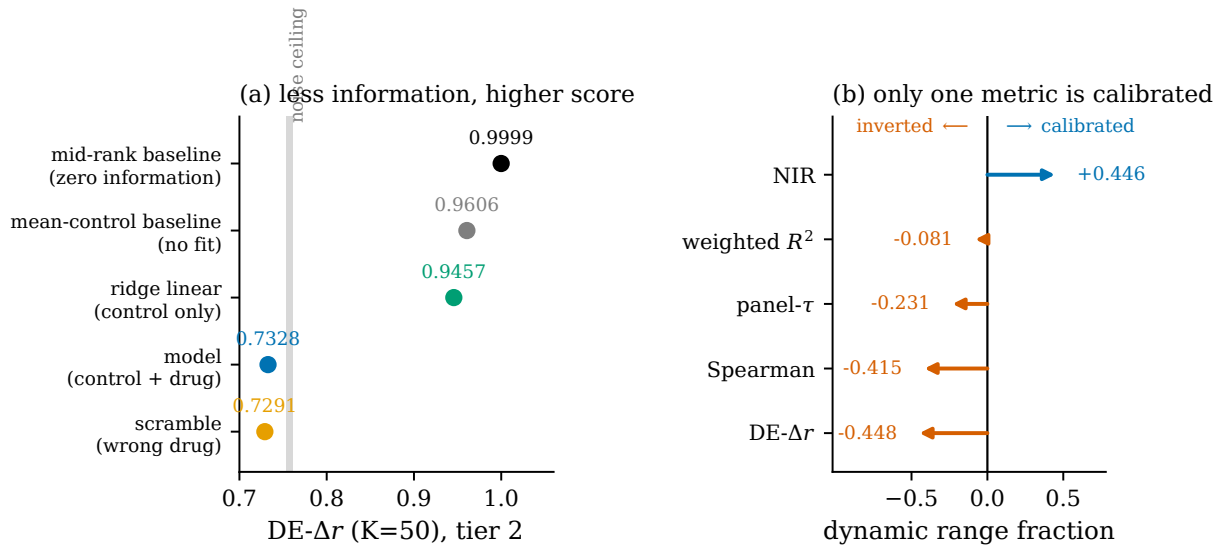


Figure 4: **Under the standard metric, less information yields a higher score.** (a) A predictor that places every gene at the middle rank — carrying nothing about drug, cell or biology — scores $\text{DE-}\Delta r = 0.9999$, above the two-real-cell noise ceiling and above both the model and its own scramble. The inverted ordering is the signature of an artifact rather than of skill, and its cause is control regression-to-the-mean. Tier 2, $n = 400$ conditions. No intervals are drawn: the stored ones are normal approximations rather than the cell-line-clustered bootstraps used everywhere else in this thesis, and mixing the two conventions on one axis would misrepresent both. (b) Grading the *metrics* rather than the models. The displacement drawn is the Dynamic Range Fraction, on a scale where 0 is the drug-agnostic baseline and 1 a perfect predictor, so a negative value means the metric ranks an uninformative baseline *above* the noise ceiling. Only NIR points forward. 655 drugs over 61 cell-line $\times$ plate groups, within plate. `weighted_r2` carries no directional claim — its sign tracks cell count, and it is approximately neutral.

Instrument	Value	Reading
Forced-choice grading	0.48 ($\approx$ scramble; ceiling 0.67–0.83)	chance
Output invariance (gap)	0.000 [−0.019, +0.016]	null
Scramble DE- Δr , tier 1	0.740 real vs 0.739 scrambled	identical
Scramble DE- Δr , tier 2	0.733 real vs 0.729 scrambled	identical
Δ_{scr} (identifiable subset)	+0.014 [−0.016, +0.042]	null

Table 6: Four independent instruments agree. Near-ceiling DE- Δr is achieved *identically* with a wrong-mechanism drug, so the score is drug-independent.

Grouping	global	cell line	MoA	MoA $\times$ cell line
tier 1 (seen conditions)	0.098	0.161	0.098	0.164
tier 2 (unseen drugs)	0.056	0.142	0.052	0.120
tier 3 (unseen combinations)	0.051	0.107	0.063	0.097

Table 7: DE- Δr of a predictor that emits the mean response of the stated group. Cell line is the only grouping that improves on a global mean; mechanism adds nothing over it, and conditioning on both is no better than cell line alone.

conditioning variable in this data is the cellular context, not the compound class — which is worth holding onto, because Section 4.3 finds precisely that asymmetry inside the model’s activations, and Section 4.10 finds that mechanism does carry signal once the generic program is subtracted. The apparent tension is a frame difference rather than a contradiction: in full-profile space the generic response dominates and swamps a mechanism effect that becomes visible only in the residual frame.

On the calibrated metric the model sits at chance and is matched by predictors that know nothing about the drug:

Predictor	NIR
ceiling (real replicate)	0.576
model	0.498
linear (drug-agnostic)	0.500
control-copy (zero drug information)	0.504
mean	0.180

Table 8: Aggregate expression-space NIR, within plate, tier 2 ($n = 606$). Chance is 0.50.

Two objections, both closed. First, *is the task winnable at all?* Only 46.2% of (drug, cell line) conditions are identifiable in principle; 27.3% of drugs are statistically inert against DMSO and 78.7% have a plate-mate closer than their own replicate. But restricting to the identifiable subset does not rescue the model: it appears to beat the linear baseline there (0.768 vs 0.531), yet a zero-drug-information control-copy matches

it exactly (0.766) and the leak-immune Δ_{scr} remains null. The model is drug-blind *even where the task is provably winnable*. The ceiling itself rises with aggregation (0.61 at 10 cells, 0.76 at 40, 0.85 at 120), so the signal is real and noise-limited.

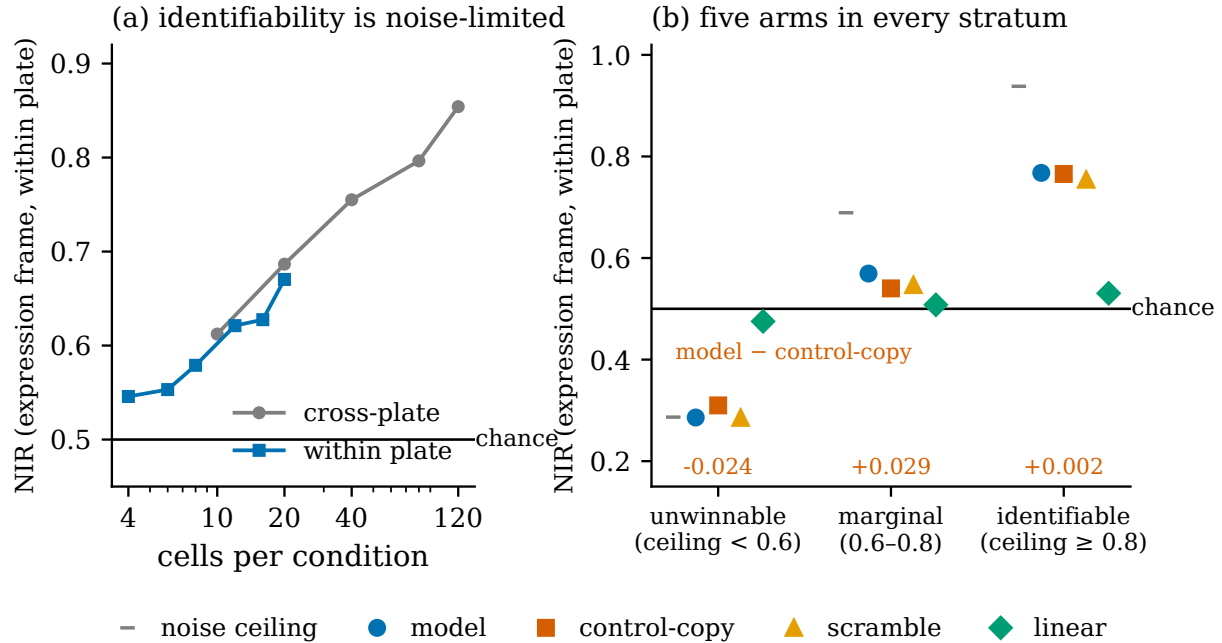


Figure 5: **The task is winnable where the noise ceiling is high — and precisely there, a predictor containing no drug information matches the model.** (a) The ceiling rises steadily with the number of cells per condition, cross-plate and within plate, so identifiability is noise-limited rather than absent: the drug signal is real and emerges under aggregation. (b) Conditions binned by their own ceiling, with five arms in every stratum. On the identifiable stratum ($n = 204$) the model reaches 0.768 — and a control-copy baseline that simply reproduces the untreated cell, and therefore knows nothing whatever about the drug, reaches 0.766. The paired contrast beneath each stratum is +0.002 $[-0.026, +0.028]$ there, clustered over cell lines. Showing the model alone across these strata would make the visual claim that it does better on winnable drugs; the control arm is what shows that it does not.

A third objection, and the one that most needs closing. Cell sentences keep only the rank order of genes and discard expression magnitude. It is natural to suspect that this is the whole story — that the drug lives in the magnitudes and the representation throws it away — and if that were so, the diagnosis in Section 4.5 would be attacking the wrong thing. The objection is testable without any model, by asking whether *true* expression discriminates drugs any better than rank does.

The representation is exonerated. At single-cell resolution true expression discriminates drugs at 0.53, which is where rank sits too; the drug is not hiding in the magnitudes. What recovers it is aggregation — 0.79 at fifteen cells under the control-referenced shift — and that is a property of the noise level rather than of the encoding. This matters twice over. It rules out the simplest alternative explanation for everything

Similarity on true normalised expression	single cell	pseudobulk of 15
cosine on expression	0.529 [0.515, 0.542]	0.723 [0.705, 0.741]
Pearson on expression	0.530 [0.517, 0.543]	0.729 [0.710, 0.746]
Spearman on expression	0.531 [0.518, 0.543]	0.627 [0.609, 0.645]
cosine on the control-referenced shift	0.542 [0.527, 0.556]	0.793 [0.775, 0.809]

Table 9: Same-drug versus different-drug forced-choice discrimination on true expression, 30 cell lines, chance 0.50. At single-cell resolution every similarity function sits within 0.04 of chance — the same near-chance regime as rank. Aggregation, not representation, is what recovers the signal.

that follows, and it means the repair in Section 4.5 cannot be read as “restoring the magnitudes”: that repair keeps rank throughout and changes only *what is being ranked*.

The same measurement bounds the task from below. At single-cell resolution the same-drug versus different-drug gap is +0.002 — different-drug agreement already sits at the replicate ceiling — and a mechanism-of-action positive control fails to separate. Whatever a model can learn here, it cannot be learned from a single cell.

Second, *does the scramble swap in similar drugs?* The swap-distance sweep resolves this: Δ_{scr} is null in every distance bin including the far tail (−0.019), across a 2.3× spread in swap distance, with a trend correlation of +0.051. Telling the model it is a drug whose true response is maximally different still does not change the output.

4.3 Where the blindness lives

A null result invites two readings, and they call for opposite responses: either the model cannot do the task, or the task cannot be done. The previous section closed the second — the ceiling rises with aggregation, a third of conditions are identifiable in principle, and the model is at chance even on those. So the failure is the model’s, and the useful question becomes *where* in the model it happens. That question cannot be answered behaviourally; it needs the weights.

Drug identity *is* in the representation: linear probes recover it at 82% (layer 9) and 76% (layer 16) against a chance rate of $\approx 8\%$. With cell line, plate and dose projected out, decodability *rises* with depth, from 0.354 at layer 2 to 0.883 at layer 16. Yet the purified drug subspace’s variance share stays flat at ≈ 0.03 , against ≈ 0.6 for cell line.

A confound in the swap’s control, found late and stated here rather than buried. [TODO: rerun in flight; the numbers below are from the confounded control and must not be quoted until it lands] The injected drug-B component lies entirely *within* the purified subspace — at most ten dimensions — while the matched-norm random vector was drawn from the *full* 2048-dimensional residual stream, and the hook ablates that subspace *before* adding. Roughly 99.5% of the random vector’s energy

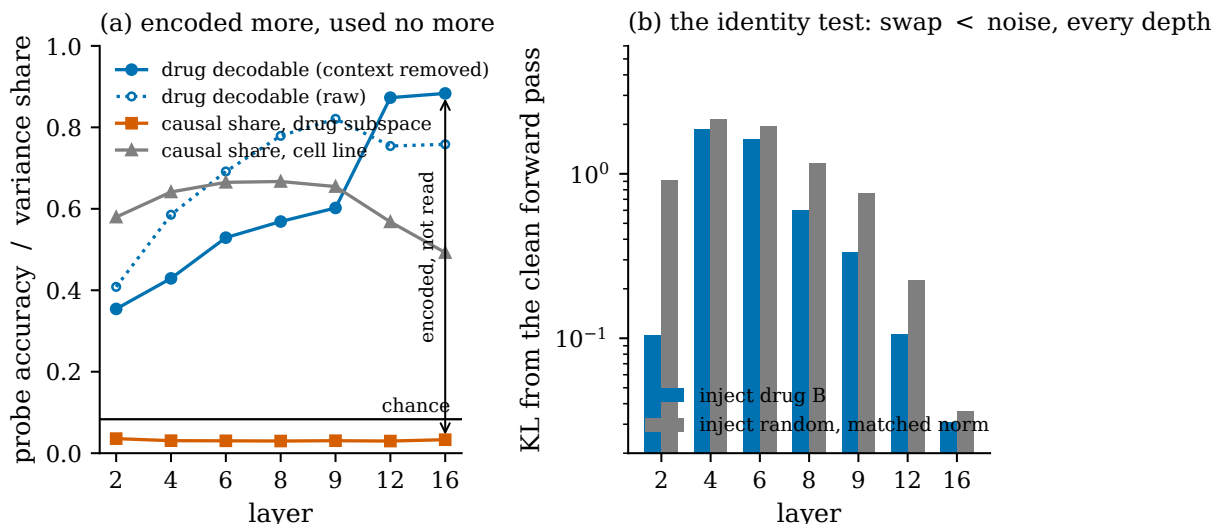


Figure 6: **The model computes an increasingly explicit drug representation and sets it aside.** (a) With cell line, plate and dose projected out, drug decodability rises monotonically with depth, from 0.354 at layer 2 to 0.883 at layer 16, against a chance rate of 0.083; the causal variance share of that same purified subspace stays flat near 0.03, against roughly 0.6 for cell line. The raw probe is drawn too and behaves differently — it peaks at layer 9 and declines — because the two series are easily conflated and the dissociation rests on the context-removed one. (b) The identity test, on a log axis because late-layer divergences are small for everything. Overwriting drug A’s code with drug B’s, inside the purified subspace, perturbs the output *less* than injecting a random vector of identical norm — at all seven depths. $n = 60$ prompts, drug means estimated on a held-out half. No dispersion is drawn on the swap: the probe reduces the per-prompt distributions to two means before storing them, so none exists, and the evidence is that seven independent depths fall the same way.

Layer	Decode (raw)	Decode (context removed)	Ablate pure drug (KL)	Random matched dim	Ratio	Swap vs matched noise
2	0.408	0.354	0.155	0.012	12.7×	0.104 < 0.917
4	0.585	0.429	1.563	0.084	18.6×	1.855 < 2.156
6	0.692	0.529	1.435	0.251	5.7×	1.618 < 1.958
8	0.779	0.569	0.596	0.042	14.1×	0.599 < 1.163
9	0.821	0.602	0.418	0.100	4.2×	0.333 < 0.761
12	0.754	0.873	0.133	0.035	3.8×	0.105 < 0.227
16	0.758	0.883	0.015	0.004	3.6×	0.031 < 0.036

Table 10: All seven layers pass the removal gate (88–95% of above-chance drug signal removed); chance is 0.083. Two decode series are shown because they behave differently and the distinction matters: the *raw* probe peaks at layer 9 and then declines, whereas with cell line, plate and dose projected out decodability rises monotonically to layer 16. The swap is the identity test — drug-B injection against a random vector of identical norm, both with drug means estimated on a held-out half — and it is null at every depth. Swap values carry no dispersion: the probe reduces the per-prompt distributions to two means before storing them.

therefore landed in directions that were never ablated, including the cell-line directions this same experiment measures at ≈ 0.6 variance share against the drug subspace’s 0.03, and up to $11\times$ its ablation KL. “The swap moves the output less than matched noise” is precisely what that arrangement predicts, whatever the readout actually does: the control matched the norm but not the subspace.

The corrected control randomises the *direction within* the slab at fixed subspace and fixed norm, reports a paired bootstrap interval rather than two bare means, and gates on the purified subspace rather than the raw one. What is unaffected is everything that does not rest on the swap: the ablation ratios, whose null was already a matched-*dimension* random subspace and is correctly constructed; the variance shares; and the decodability-versus-depth dissociation. What is affected is the direction-blindness claim specifically, which rests on the swap alone.

The headline is the swap. Overwriting drug A’s code with drug B’s, inside the purified subspace, perturbs the output *less* than matched-norm noise, at all seven depths. The readout responds to the *magnitude* of activity in the drug region, not to *which* drug it encodes. Held-out mean estimation reproduces this exactly (layer 2: 0.104 vs 0.106; layer 9: 0.333 vs 0.334; layer 16: 0.031 vs 0.031), ruling out estimation noise as the cause.

Validity of the null, and how much of it survives. Two pillars stand and one does not. The removal gate passes at every layer, so the nulls are not broken hooks. The cell-line positive control carries $\approx 20\times$ the purified drug subspace’s variance share (0.6 against 0.03) and up to $11\times$ its ablation KL, so the instrument does register a strong

effect where one exists — but that KL ratio is not uniform, and at layer 12 it falls to $1.02\times$, meaning the instrument has essentially no headroom at that depth and no claim should rest on it. The third pillar, that the swap is translation-clean by matched norm, does not survive: as described above, the norm was matched but the subspace was not.

Reading. The clean claim is not that the drug lives in a dead subspace. It is that **the generation readout is drug-agnostic within an increasingly explicit drug representation**. That pattern is what the capacity-allocation account of Section 2.6 predicts [23]: cell-line identity is consulted on every forward pass, because emitting any plausible profile requires it, and so earns dedicated causal weight; drug→gene response is needed only for context-dependent composition, and can be encoded ever more explicitly without the readout ever querying *which* drug it names. The reading is ours, not theirs — we are applying a claim about how capacity is allocated in language models to a case it was not derived from — and it earns its place here by being falsifiable rather than by being borrowed. It predicts the next result: if the defect is a readout that ignores direction, no change to the target can touch it.

4.4 The ε -ladder: no target-side fix works

The mechanism makes a falsifiable prediction, and this section is its test. If the defect is a readout that feels the magnitude of activity in the drug region but not its direction, then improving what sits *in* that region cannot help — and improving the training target is exactly that kind of intervention. The obvious objection is that one failed target construction proves nothing, so rather than try another we swept the family: entropic regularisation gives a one-parameter path whose two endpoints are the two constructions already tested independently, which turns a sequence of attempts into a controlled sweep.

Target	ε	Δ_{scr}	Verdict
single cell (real treated cell)	≈ 0	+0.014 [−0.016, +0.042]	null
OT barycentric (state-matched)	0.5	−0.001 [−0.018, +0.016]	null
consensus (global mean)	$\rightarrow \infty$	−0.010 [−0.022, +0.003]	null

Table 11: The regularisation parameter interpolates between the two target constructions tested independently, and no point on the ladder lifts the model out of the near-null regime. These figures come from a *random-partner* scramble; Section 4.5 introduces a sharper stratified test under which the optimal-transport arm does show a weak effect (+0.0263 [+0.0069, +0.0476]), roughly five-fold below what re-encoding the target achieves. The consensus endpoint is additionally the ladder’s weakest-evidenced point: that arm was stopped at roughly 60% of an epoch and its tier-1 NIR was unscorable, so the negative rests on the single-cell and optimal-transport endpoints.

Training was not inert: the consensus model stops echoing the control (0.653 against

a control-copy of 0.766, where the single-cell model sits *on* the control at 0.771) and its predictions are not mode-collapsed. But the variation is uncorrelated with real drug identity (Mantel $\approx +0.03$). Not collapse — wrong structure.

As predicted by Section 4.3, denoising changes *what is in* the target while the defect is the readout ignoring direction within it.

4.5 The repair: it was the tokenisation

Every target-side intervention changed *what the target was* and none changed *what its tokens encode*. That distinction turns out to be the whole story, and it is measurable rather than rhetorical: the question is not how good the target is but what fraction of its tokens carries drug identity at all.

Representation	Inter-drug top-200 tokens differing	Replicate retrieval ceiling
full profile (the standard target)	34.6 / 200 (17%)	0.742
shift (treated – control)	111.4 (56%)	0.742
residual (drug-specific)	118.2 / 200 (59%)	0.7432

Table 12: 83% of the standard target’s tokens are identical between any two drugs in the same context. The residual row is quoted at **cell-line** scope, which is how the training targets were actually built; at plate scope it reads 120.3 and 0.7467. Note the retrieval ceiling is *unchanged* across all three rows: the information was always present, and the tokenisation discarded it.

Because a cell sentence ranks genes by expression, cross-entropy allocates roughly 1% of its gradient to drug identity. This explains why every change to target *content* failed identically. Re-encoding the target as the drug-specific residual raises the differing fraction to 60% and produces the first non-null result in the project:

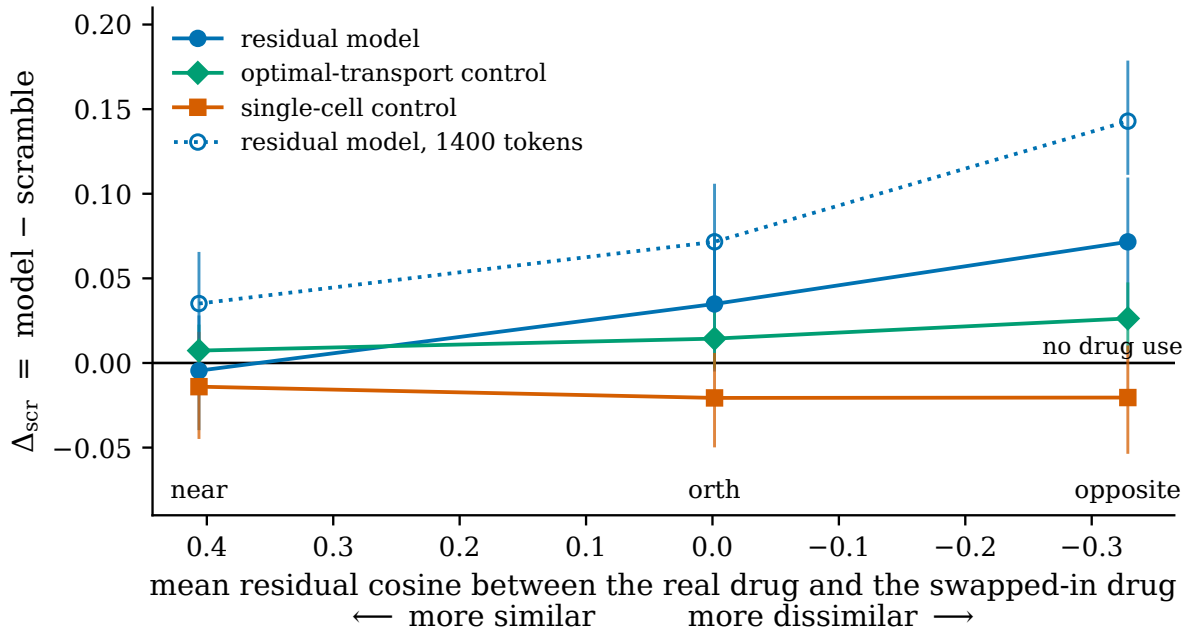
Swap stratum	Residual model	Single-cell (control)
near (cos = +0.41)	+0.0351 [+0.0061, +0.0656]	−0.0014
orth (cos $\approx$ 0)	+0.0716 [+0.0371, +0.1059]	−0.0006
opposite (cos = −0.33)	+0.1429 [+0.1112, +0.1787]	−0.0084 [−0.0394, +0.0234]
Gradient with dissimilarity	yes (all three clear zero)	no (flat)

Table 13: Stratified scramble. The monotone gradient is itself the evidence: a metric artifact would not order itself by swap dissimilarity. Control columns are quoted at a matched 1400-token generation budget; see below.

A budget confound, checked rather than argued away. The residual model’s figures came from a run at 1400 generation tokens while both control arms had originally run at 600 — and since the token budget alone doubles the measured effect for the residual

model, that is a real asymmetry, biased in the direction that flatters our conclusion. There was a plausible argument that it did not matter, namely that a cell sentence is roughly 123 gene symbols against a residual signature’s 201, so the cap binds far less on a control. But the generation completion rates were logged and never persisted, so the argument could not be checked against the artifacts, and an unverifiable argument in favour of one’s own result is worth very little.

Re-running both controls at 1400 settles it. The single-cell control moves from $-0.014/-0.021/-0.020$ to $-0.001/-0.001/-0.008$ — flat and null at every stratum, as before — and the optimal-transport control from $+0.007/+0.014/+0.026$ to $-0.007/+0.016/+0.025 [+0.008,+0.042]$. Both are unchanged in substance. The comparison is now matched and the conclusion is unaffected.



residual frame. Intervals are 95% bootstrap, clustered over cell lines; $n = 250$ conditions per point.

Figure 7: **The residual model’s advantage over its own scramble grows with how dissimilar the swapped-in drug’s response is; a control scored through the identical pipeline stays flat and null.** The evidence is the gradient, not any single point — a metric artifact would not order itself by swap dissimilarity. The horizontal axis is the measured mean residual cosine between the real and swapped-in drug, which is byte-identical across all four runs, so the positions are exact. **The three solid series are at a matched 600-token generation budget:** plotting the residual model’s 1400-token run against 600-token controls would compare a treatment arm at more than twice its controls’ budget, since the budget alone doubles the effect. That run is drawn separately as the dotted overlay. Intervals are 95% bootstrap clustered over cell lines, $n = 250$ conditions per point. The **near** stratum is the weakest test by construction: where the swapped-in drug’s true response is similar, an unchanged output is correct rather than blind.

Three further checks agree, and only the residual model passes all three: the correlation

between condition reproducibility and model NIR is $+0.144$ (positive: the model does better where the drug effect is real) against -0.117 and -0.216 for the controls; prediction diversity is highest for the residual model; and while $\cos(\text{prediction}, \text{own truth})$ exceeds $\cos(\text{prediction}, \text{other drugs})$ for all three arms, the residual model’s margin ($+0.084$) is roughly twice the optimal-transport arm’s ($+0.049$) and three times the single-cell arm’s ($+0.026$).

A methodological note. The effect was first measured at a 600-token generation cap, which silently truncated 26% of generations — a signature is ≈ 201 gene symbols at 2–4 BPE tokens each — and *halved* it ($+0.0716$ against the correct $+0.1429$). For any generation-based evaluation, log the completion rate.

A retraction. The OT arm was originally reported as achieving nothing. Under the sharper stratified scramble it scores $+0.0263$ [$+0.0069$, $+0.0476$] with a monotone gradient: it does use the drug, weakly. The original null was an underpowered instrument, not an absence of signal. The ε -ladder conclusion stands, but “OT achieved nothing” is withdrawn.

4.6 The drug use generalises

A model that reads the drug token has cleared a low bar, because memorisation requires reading it too. Everything scored above came from conditions the model trained on, so the result so far shows that the token is consulted — not that anything transferable was learned. Separating those two requires withholding data, and it requires withholding it at the right granularity.

Split	n (cell lines)	model NIR	Δ_{scr} (opposite swap)	Verdict
train	200 (40)	0.657	$+0.0898$ [$+0.0530$, $+0.1269$]	uses drug
unseen_combo	250 (38)	0.650	$+0.1002$ [$+0.0661$, $+0.1368$]	transfer
unseen_drug	120 (36)	0.586	$+0.0195$ [-0.0452 , $+0.0860$]	null (by design)

Table 14: Three-way holdout keyed at the (drug, cell line) level. Validity clean first: 9,120 generations, [DOWN] present in 93%, 98.9% valid panel genes, 1.4% duplicates.

There is no memorisation premium. Held-out pairings score *as well as* trained ones ($+0.1002$ vs $+0.0898$, intervals almost entirely overlapping). The model did not memorise pairings; it learned a drug representation that applies in cell lines where it never saw that drug. The `unseen_drug` arm behaving as designed — same model, same instrument, drug token carrying no learned information, no gap — is what licenses attributing the transfer to having seen the drug.

Two instrument defects, both of which would have concealed this. A uniform condition sampler reproduces the pool’s split proportions, so 250 constructed held-out combinations yielded only 33 scored ones; and a strict-inequality NIR scored the zero vector (“predict the generic response”) at 0.000 rather than 0.500. Both are corrected in Section 3, and after correction the drug-agnostic baselines land where they must: **generic** exactly 0.500, **control_copy** 0.478, **random** 0.502. The residual frame is therefore *demonstrably* leak-proof rather than assumed to be.

4.7 But a lookup table wins

Transfer without memorisation is a real property, but it says nothing about quality: a model can apply a learned signature in a new context and still apply it badly. The pre-registered bar was therefore never “beats chance” — residual targets are plausibly memorisable, and a predictor that simply looks up what this drug did elsewhere needs no model at all. That predictor is the comparison that matters.

Arm	NIR	model – arm (clustered CI)
ceiling (within-condition replicate)	0.968	—
drug_lookup	0.963	−0.3245 [−0.3539, −0.2975] loses
drug_lookup_1	0.832	−0.1936 [−0.2285, −0.1612] loses
model	0.639	—
moa_lookup ($n = 143$)	0.529	+0.1042 [+0.0144, +0.1882] wins
generic	0.500	+0.1388 [+0.1086, +0.1650] wins
random	0.502	—
control_copy	0.478	+0.1613 [+0.1314, +0.1907] wins

Table 15: The baseline ladder, $n = 570$. On the held-out split specifically, model – **drug_lookup** is −0.3170 and model – **drug_lookup_1** is −0.1825.

The pre-registered interpretation, written before the result, anticipated three branches; the middle one fired. The model reads and transfers the drug, but recovers only a small fraction of its signature. That it beats **moa_lookup** rules out the explanation that it is simply reading the **Mechanism:** field of the prompt — though at 0.529 that is a low bar.

This is DrEval’s finding, in transcriptomic space. DrEval reports that deep learning models barely outperform a naive predictor of mean drug and cell-line effects [13]. Their **NaiveMeanEffectsPredictor** is $\hat{y}_{ij} = \mu_i^c + \mu_j^d - \mu$; our **drug_lookup** is the $\beta(d)$ term of that same additive model, measured in a frame where the generic(c) term has already been subtracted. The finding replicates and sharpens: on a 946-gene transcriptomic response rather than a scalar potency, and with the comparison made against a proper within-condition replicate ceiling, the additive drug main effect reaches 0.963 of a 0.968 ceiling while a 1B-parameter model reaches 0.639.

Note also that this result lands in the split where DrEval specifically warns the drug-average baseline will look strong: `unseen_combo` is Leave-Pairs-Out, and DrEval reports that good LPO performance is reachable by predicting each drug’s training average [18]. We take that warning as the hypothesis rather than the caveat, and measure it. It holds.

A retraction, made before it entered the record. It is tempting to read ceiling – `drug_lookup` = +0.005 as “the modelling target has vanished”. That inference is not supported, for three reasons that all favour the lookup: the ceiling is scored against a *half-sample* truth while every baseline is scored against the *full-sample* truth; `drug_lookup` averages ≈ 38 conditions against the ceiling’s single noisy half; and NIR near 0.96 is compressive, collapsing a wide range of cosines into a few thousandths.

4.8 Every arm in one place

The results so far are distributed across two measurement frames and a dozen tables, and a reader assembling them has to keep track of which numbers may be compared with which. Table 16 collects them.

The two frames are *not* interchangeable and are separated deliberately. The expression frame scores a predicted cell profile against other drugs on the same plate; the residual frame scores a predicted drug-specific signature against other drugs in the same cell line, after the generic program has been subtracted. Both have chance at 0.50 by construction, and that coincidence is the trap: the residual frame’s ceiling is 0.968 where the expression frame’s is 0.576, so a model scoring 0.639 in one and 0.498 in the other has not improved — it is being measured against a different question.

Read down the residual block and the thesis’s shape is visible in one object. A replicate of the experiment reaches 0.968; a lookup table reaches 0.963; the model reaches 0.639; the same model told the wrong drug reaches 0.559. The distance between the last two is the drug effect this work recovered, and the distance between the model and the lookup is what it did not.

4.9 How much is left? The transfer coefficient

Sections 4.6 and 4.7 leave two numbers pointing opposite ways, and NIR cannot adjudicate between them. `drug_lookup_1` sits 0.136 below the ceiling despite using *more* cells than the ceiling’s half-sample, arguing for real cell-line-specific structure; yet averaging many lines recovers to 0.963, and averaging removes only *noise*, never the systematic miss of the target line’s own interaction. The question is therefore posed in cosine space with the noise divided out (Section 3.10).

Frame	Arm	NIR	What it knows
expression (within plate, tier 2, $n = 606$)	noise ceiling	0.576	a real replicate of this condition
	control-copy	0.504	nothing — it reproduces the untreated cell
	linear map	0.500	the control cell, no drug
	model	0.498	control cell + drug
	global mean	0.180	nothing
residual (cell-line sets, $n = 570$)	noise ceiling	0.968	a real replicate of this condition
	drug lookup	0.963	this drug’s signature from other cell lines
	drug lookup, one line	0.832	the same, from a single other cell line
	model	0.639	control cell + drug
	scramble (opposite)	0.559	control cell + the <i>wrong</i> drug
	mechanism lookup	0.529	other drugs sharing this drug’s mechanism
	random signature	0.502	nothing
	generic response	0.500	nothing (the zero vector, by construction)

Table 16: Every arm on the calibrated metric, by frame. Chance is 0.50 in both. The residual-frame figures come from a single evaluation run; a replication at further seeds is in progress, and Section 5 records that the intervals accompanying these point estimates capture between-cell-line variance but not generation variance.

A false alarm, and the control-design lesson it produced. The first run compared $\mathcal{T}$ against a negative control of *different drugs on the same plate*, which returned +0.485 against $\mathcal{T} = 0.511$, and was correctly declared void. But that control differs from $\mathcal{T}$ in **two** ways at once — different drug *and* same cell line. Plate structure surviving a cell-line-scoped generic inflates *same-plate* comparisons while leaving *cross-line* comparisons untouched, so it destroyed the control’s usefulness without ever touching $\mathcal{T}$. A *structure-matched* control — different drug, **different cell line**, breaking only the drug match — resolves it, reading +0.000 at both plate and cell-line scope and +0.010 with the generic projected out. The episode is worth reporting rather than hiding: a negative control that differs from the estimand in more than one way cannot bound it.

Configuration	$\mathcal{T}$	same-plate null	matched null	dose	share
plate-scoped generic	0.557 [0.513, 0.601]	−0.018	+0.000	0.703	
plate + generic projected out	0.545 [0.503, 0.585]	−0.019	+0.010	0.682	
cell-line-scoped (production)	0.517 [0.474, 0.560]	+0.478	+0.000	0.484	

Table 17: The transfer coefficient. Against a simulated $\kappa = 0$ null of 1.001, the shortfall is 0.444–0.484. The matched negative control is ≈ 0 throughout.

Answer: roughly 45% of the drug-specific residual variance is drug×cell-line interaction , robust to the generic’s scope, to projecting the generic direction out, and to control splitting. The per-drug constant $\beta(d)$ — which `drug_lookup` reproduces at

0.963 — is therefore *not* the whole signal. There is a conditional target worth about half the drug-specific variance that no lookup can structurally reach, and that the model does not reach either.

Three internal checks agree. **Dose ordering:** at plate scope, same-drug/same-line/different-dose transfer is 0.703, *above* cross-cell-line 0.557 — changing the dose costs less than changing the cell line, the biologically correct ordering; at cell-line scope the ordering inverts. **Control-split bias:** at plate scope the shared- and split-control builds agree to -0.000 , against $+0.254$ at cell-line scope, so that discrepancy was plate contamination rather than the split. **Orthogonality:** $(\|r\|/\|s\|)^2 + (\|\text{generic}\|/\|s\|)^2 = 1.002$, so the two components are essentially orthogonal and the fractions below are genuine variance shares.

Is that interaction learnable, or merely present? $\mathcal{T}$ says how *large* κ is; it says nothing about whether anything could learn it. If a cell line modulated drug responses in a consistent direction, that direction would be estimable from the line’s own cells and a conditional model would have a concrete target. Writing $\kappa(d, c) = r(d, c) - \hat{\beta}(d)$ with $\hat{\beta}$ estimated leave-one-cell-line-out — necessary, since a mean containing $r(d, c)$ would bias the signal arm downward by construction — and disattenuating by κ ’s own split-half reliability:

Configuration	same cell line	different cell lines	excess within-line
plate-scoped generic	-0.004 $[-0.011, +0.002]$	$+0.003$	-0.007
cell-line-scoped	-0.003 $[-0.021, +0.015]$	$+0.002$	-0.005

Table 18: No cell line bends different drugs in a consistent direction. 4,000 pairs per arm; the signal interval at plate scope excludes anything above $+0.002$, so this is a real null rather than an underpowered one. The cell-line-scoped row is wider ($[-0.021, +0.015]$) and carries the claim less strongly.

The interaction is idiosyncratic: real, large, and specific to each (drug, cell line) pairing. This resolves the tension the previous section could not. A lookup reaching 0.963 against a 0.968 ceiling is *not* evidence that κ is small — it is 45% of the variance. It is that κ is *unpredictable*, so it behaves as irreducible noise for any predictor, and the lookup is therefore already at the *achievable* ceiling. Nothing beats it on seen drugs.

What a lookup structurally cannot do is handle a drug it has never seen — and Section 4.10 shows that regime has live channels. The model’s opportunity is not the interaction; it is the unseen drug.

Scope of the whole thesis. At plate scope $\|r\|/\|s\| = 0.786$ and $\|\text{generic}\|/\|s\| = 0.620$: **62% of the perturbation response variance is drug-specific**, and 38% is the

generic program plus batch. Every claim in this thesis is scoped by this number, and it has not previously been computed for this task.

A retraction this forces. The justification for estimating the generic per cell line rather than per plate was a reproducibility comparison, 62% against 19%. That was measured with *shared* control cells, which inflate cell-line reproducibility far more than plate reproducibility. Measured with split controls the two are comparable (20% plate, 16% cell line), so the argument does not favour cell-line scope. Combined with the dose-ordering inversion and the control-split bias, the plate-scoped build is the better one — and the residual training targets, built at cell-line scope, therefore retain batch structure. This does not affect Δ_{scr} , which is a paired contrast in which any shared component cancels, but it makes a plate-scoped rebuild the natural next arm.

4.10 Scope: what is closed by measurement

An unseen drug can only be predicted through a property it *shares* with drugs seen in training, so the question is answerable by enumerating the channels and measuring each in the same frame. For each channel we predict a drug’s residual from the mean residual of the *other* drugs in the same cell line that share the property — the construction of `moa_lookup`, generalised. Each arm is paired with a **count-matched random null** drawing the same *number* of partners at random, without which a channel scoring above chance cannot be distinguished from the fact that averaging k residuals denoises.

Channel	Coverage	n	NIR	vs. count-matched null	Verdict
protein target	54%	894	0.586	+0.0844 [+0.0592, +0.1095]	live
mechanism (MoA)	32%	1053	0.573	+0.0780 [+0.0555, +0.1027]	live
chemical structure	100%	4084	0.484	+0.0137 [−0.0013, +0.0278]	closed

Table 19: Coverage is stated first: a channel annotated on few drugs cannot be *closed* by a null result, because it was never tested. Chemistry reproduces the structure gate in this frame; both knowledge channels clear their nulls.

This overturns a pre-registration made in this project. Two design documents concluded that unseen-drug generalisation was out of reach, and both inferred it from the structure gate — that is, from Morgan fingerprints and a molecular language model. But the drug-knowledge specification written earlier in the same project ranks raw structure the *lowest*-value channel and protein targets “the key feature”, on the reasoning that transcriptional response is driven by which protein a compound hits, something structure encodes only indirectly. The pre-registration generalised from the weakest channel to the strongest, and it was wrong. Tahoe’s own metadata carries protein-target annotations

for 264 drugs — 280 distinct symbols, 550 edges — which no analysis in this project had read.

The correct statement is therefore narrower and more useful than the one we started with: *chemical structure does not predict response, measured twice in two frames; protein target and mechanism do*. Unseen-drug generalisation is reachable through drug-side *knowledge*, and not through chemistry.

Three caveats bound it. Coverage is 54% and 32%, so this is measured on the annotated subset. The absolute NIR of roughly 0.58 is far below the 0.963 a lookup achieves for a *seen* drug, so the reachable ceiling in this regime is much lower. And target and MoA overlap, so whether target adds anything *over* MoA is a further question this measurement does not settle. Settling it would need a leave-one-mechanism-out split rather than the leave-one-drug-out split used here, since a held-out drug’s mechanism class is generally still represented in training — and, as the next section shows, there is a prior reason to expect the answer not to matter much.

4.11 But the model does not read the channel it already has

The natural next step from the preceding section writes itself: append the protein targets to the prompt, retrain, and see whether unseen-drug performance rises. That arm assumes the bottleneck is *information availability*. Section 4.3 says it is the readout. Before spending two GPU-weeks discovering which, note that three facts already sit uneasily together — the mechanism channel carries real signal, the prompt has always contained a **Mechanism:** field, and the model’s unseen-drug gap is null — and that the only thing currently reconciling them is that the unseen-drug arm is underpowered. That is true, and it is a dodge.

The scramble can settle it directly, because it has until now replaced the drug name *and* the mechanism together. Decomposing it into single-span swaps costs one extra pair of generation arms:

Swap (opposite-signature partner)		Δ_{scr}	n	
drug name only (mechanism kept)	+0.0809	[+0.0592, +0.1010]	570	read
mechanism only (name kept)	+0.0091	[−0.0174, +0.0383]	325	not detected
both (the standard arm)	+0.1024	[+0.0827, +0.1236]	570	read

Table 20: Each arm swaps exactly one span of the instruction and leaves every other byte identical. The mechanism arm carries fewer conditions because a partner sharing the original’s mechanism would produce a prompt identical to the model’s own — a gap of zero by construction — and those conditions are dropped rather than scored.

The model reads the name and does not measurably read the mechanism. The two are roughly additive, with the name carrying about 79% of the combined effect.

This is not an underpowered null: an effect the size of the name’s would clear an interval of half-width 0.028 with room to spare, and what is observed is +0.009.

So the model is handed a field that demonstrably carries drug-transferable signal and does not use it. For unseen drugs the bottleneck is not what the model is told — it is the same readout failure Section 4.3 localises, surfacing in the one place where it costs the most. And the arm that would have followed from the previous section is thereby closed before being built: appending protein targets to a prompt whose mechanism field is already ignored would change nothing. A single five-hour evaluation retired roughly two GPU-weeks of work, which is an argument for decomposing an ablation before scaling one.

5 Limitations and Future Research Directions

5.1 Limitations of scope

Drug coverage. The residual analysis is built on a cache of 620,564 cells covering 106 drugs across 50 cell lines, sampled from Tahoe’s shards — roughly a tenth of the atlas’s compound set. This is ample to test whether the model uses the drug, and to measure transfer, but it is not a population estimate. Any claim about the *distribution* of drug behaviours should be read against the broader characterisation, which used 6,628 conditions.

Only reproducible conditions. The residual arm retains the 62% of conditions whose residual reproduces across a half-split ($\cos > 0.2$). This is the right choice — training on irreproducible residuals is training on noise — but it means results describe the winnable fraction of the task, not all of it. The filter might be expected to select *toward* context-independent conditions and so bias the transfer coefficient upward. Measured, it does the opposite: the unfiltered estimate is 0.598 against the filtered 0.557. The filtered value is the one quoted, because disattenuation is unstable at the very low reliabilities the filter removes.

The intervals omit generation variance. Every confidence interval in this thesis is a clustered bootstrap that resamples *cell lines* while treating each condition’s score as fixed. It therefore captures between-cell-line variance and not the variance introduced by sampling the model’s output. With four generations per arm at temperature 0.8 the second source is not negligible: two evaluations of the same checkpoint on the same conditions, differing only in that one carried two additional scramble arms and so consumed the sampling stream differently, returned Δ_{scr} on the held-out-drug split as +0.0195 [−0.045, +0.086] and +0.1114 [+0.051, +0.173] respectively. The headline arms were stable across the pair, and the held-out-drug arm — the smallest, at $n = 120$ — was not. **[TODO: seed replication in flight; report the across-seed spread beside the within-seed interval and state which sources each captures]** Until that lands the held-out-drug arm should be read as underdetermined rather than as a null, which weakens its use as the control licensing the transfer result.

Pseudobulk resolution. Residuals are condition-level pseudobulks over ≥ 40 treated cells. The single-cell heterogeneity of the response is not modelled in this frame, and the project’s own earlier results show why: at single-cell resolution the drug moves a cell less than noise does ($\text{SNR} \approx 0.75$, median 0 differentially expressed genes at FDR $q < 0.05$).

Cross-plate absolutes. The residual-space scores are computed with comparison sets grouped by cell line rather than by plate. The *difference* Δ_{scr} is leak-immune, because both arms share the same control cell, but the absolute NIR values in that frame are not directly comparable to the within-plate tables elsewhere. **[TODO: rescore the generation-free arms within plate — this costs no GPU time]**

Frame coverage. The drug-specific residual is $\|r\|/\|s\| = 0.786$ of the response, against 0.620 for the generic program and batch, and the two are essentially orthogonal ($0.786^2 + 0.620^2 = 1.002$). So this frame covers 62% of the response *variance* and every result here should be read within that bound — a real but not total share. The remaining 38% is precisely the component the field’s absolute-accuracy metrics are dominated by, which is why a drug-agnostic predictor scores so well on them.

5.2 Limitations of the unseen-drug evaluation

Two defects, both of which must be stated rather than papered over.

It is not a clean held-out-drug design. The prompt contains `Mechanism: {moa}`, so holding out a drug still hands the model the held-out drug’s class label. The correct design is leave-one-*mechanism*-out. The leak is probably small — `moa_lookup` scores 0.529, so the mechanism label carries little about the response — but the design is wrong and no unseen-drug claim should be made until it is fixed.

It is underpowered by construction. The confidence interval half-width on the `unseen_drug` gap is $\approx \pm 0.087$, against a maximum effect available through the mechanism channel of ≈ 0.033 . The arm therefore *cannot* detect the largest effect it could plausibly be measuring. Its null is a statement about power, not about absence, and is reported as such.

5.3 Directions

The arm this work closed rather than opened. The obvious continuation of Section 4.10 is a channel-conditioned prompt: append protein targets to the instruction, retrain, and evaluate on a leave-one-mechanism-out split. We do not recommend it, and the reason is measured rather than argued. Section 4.11 shows the model reads the drug *name* (+0.081 [+0.059, +0.101]) and does not measurably read the *mechanism* field it is already given (+0.009 [−0.017, +0.038]), in an arm powered to detect an effect the size of the name’s. Adding a target field to a prompt whose mechanism field is ignored addresses information availability, and the binding constraint is the readout.

What would be worth building instead is an intervention that makes the drug *causal* rather than merely present — a learned per-drug modulation of the transformation from control to treated, so that the drug parameterises the computation instead of sitting in the residual stream as an input the readout can round away. That is a research architecture rather than a prompt change, and it is outside the scope of what remained here.

Two channel questions that do remain open. Target and mechanism overlap, so whether target adds anything *over* mechanism is unresolved — and it matters, because the prompt already exposes mechanism and the model already ignores it. Separately, the channel gate measures *similarity transfer*: an unseen drug resembling a seen drug with the same target. It does not test a *compositional rule* — that a named target’s own pathway moves, which would apply to any drug whose target is named, including one with no similar drug in training. The second is the stronger claim, it does not reduce to a lookup, and it is untested.

Unseen cell lines. A cell line never seen with any drug is a different generalisation axis from the held-out pairings tested here, and one not blocked by the structure gate: it asks whether the control cell’s state is used *compositionally* with the drug. The transfer coefficient now scopes it: with the interaction both substantial ($\approx 45\%$) and idiosyncratic, a new cell line offers no consistent direction to exploit, so the achievable target there reduces to recalling the drug’s per-drug constant — which a lookup already does at 0.963. The axis is therefore less promising than it appeared before Section 4.9 was measured.

Objectives that were considered and rejected. Recorded so they are not re-litigated:

- **Reinforcement learning with a contrastive reward.** Its mechanism — a per-sequence reward, so the token-dilution ratio never enters the gradient — attacks a bottleneck that the residual encoding has already lifted ($34.6 \rightarrow 118.2$ differing tokens); and its reward optimises similarity to the true residual, which is precisely what `drug_lookup` already achieves at 0.963. Its best case is convergence to the lookup.
- **A differentiable profile-level contrastive loss.** The natural construction, $\text{score}(g) = \sum_i P(\text{token}_i = g) w(i)$, presumes $\text{gene} \approx \text{token}$, but 86.9% of the 946 panel genes share a first sub-token under this tokeniser (mean 3.254 BPE tokens per gene, only 8 single-token). Under teacher forcing the difficulty compounds: the residual target *is* the drug signature, so the true prefix identifies the drug within a handful of symbols and the gradient vanishes exactly at the drug-blind solution the loss exists to penalise.

- **Loss re-weighting toward differentially expressed genes.** Re-weighted cross-entropy against a single target contains no negatives, so nothing in it forces the output for one drug to differ from another’s. Given that the readout is direction-blind (Section 4.3), it addresses token dilution but not the actual defect.

One model size, and the smallest one. Every result here comes from a 1B-parameter backbone. C2S-Scale releases checkpoints up to Gemma-2 at 27B, and we tested none of them, so the most predictable question an examiner can ask — does it work if the model is bigger — has no answer in this thesis.

Two things can be said about what we would expect, and they point in different directions. The bottleneck this work identifies is not capacity. It is that roughly 1% of the training gradient reaches drug identity, because the target tokens are overwhelmingly shared across drugs (Section 4.5); a larger model receives the same diluted signal from the same objective, and the fact that re-encoding the target lifted the effect while leaving capacity untouched is evidence that the constraint binds where we say it does.

The genuinely different argument for scale is not capacity but *prior knowledge*. A 27B model pretrained on far more text plausibly knows, as a fact about the world, that a named compound inhibits a named kinase — knowledge a 1B model may simply lack. That matters here because Section 4.11 finds the model reads the drug *name* but not the *mechanism* field it is already handed, and a model with real pharmacological knowledge is exactly the case where that might change. This is a sharper hypothesis than “bigger is better” and it is testable cheaply with a low-rank adaptation of a mid-sized checkpoint rather than a full fine-tune. We did not run it, and it is the first thing we would run with more time.

The starting checkpoint was never prepared for this task. We fine-tune from C2S-Scale-Pythia-1b-pt, which is pretrained on atlas data alone — CELLxGENE and the Human Cell Atlas, no perturbation corpus of any kind — and whose released capabilities do not include perturbation prediction. This cuts two ways and both should be stated. It is why the held-out splits are clean: a drug withheld here was never seen at any stage, so nothing in the pipeline can have memorised it. But it also means the failure we report is a failure of *this* starting point, and a reader may reasonably ask whether a checkpoint already tuned for perturbation prediction would behave differently. We cannot answer that from the public artifacts. The perturbation-specialised model in the same line of work is a Gemma-3 4B checkpoint that was not released, and the available C2S-Scale checkpoints are the pretrained ones. What we can say is narrower and worth separating from the broader claim: every arm in this thesis starts from the same base and differs only in the variable under test, so the *comparisons* between arms are unaffected by this choice even where the absolute level is not.

Generality. Everything here is one backbone on one atlas in one representation. Whether the readout-specificity failure is a property of cell-sentence models specifically, or of autoregressive conditional generation over transcriptomes generally, is open. The mechanistic protocol of Section 3.5 — purified subspace, removal gate, positive control, matched-norm swap — transfers unchanged to any model with an accessible residual stream, and is probably the most reusable artifact of this work.

6 Conclusions

This thesis asked whether a large language model fine-tuned on the largest available single-cell drug-perturbation atlas learns drug-specific biology. It does not — and the interesting content is in the chain of results establishing why, how far the failure can be repaired, and what bounds the repair.

The evaluation had to be fixed first. The metric the field standardises on is exploitable: a zero-information baseline placing every gene at the middle rank scores $DE-\Delta r = 1.000$, above both the noise ceiling and the model. Under a Dynamic Range Fraction analysis computed within plate, only NIR — a discrimination metric, in which the generic response cancels — is calibrated. Of the four prediction metrics tested, three are clearly inverted and the fourth is approximately neutral. This is the most portable result here, since it applies to any work scored on $DE-\Delta r$ -family metrics.

This also settles how the thesis enters the dispute of Section 2.3. Miller et al. [14] argue that reports of deep models failing to beat uninformative baselines reflect miscalibrated evaluation rather than model failure, and on genetic perturbation they support that with calibrated-metric benchmarks. We port their framework, confirm its methodological claim emphatically — metric choice dominates, and only the rank-based metric survives — and then find that on drug perturbation the model is at chance *on the metric their own analysis endorses*. That is the strongest form the sceptical result can take: it cannot be answered with the objection their paper raises. Their conclusion is not overturned so much as bounded — it does not extend from genetic perturbation to this task, and the per-drug-constant finding above suggests why.

The model is drug-blind, and the blindness is a readout failure. Four independent instruments agree that swapping the drug token changes the output by nothing. Yet drug identity is linearly decodable from the residual stream, rising to 0.88 by layer 16 with context removed, while the purified drug subspace’s causal variance share stays flat near 3%. A matched-norm swap — overwriting one drug’s code with another’s — moves the output no more than matched noise at every depth. The model computes an increasingly explicit drug representation and sets it aside. Read through representational selectivity, the asymmetry is principled: cell-line conditioning is a fixed lookup that emitting any plausible profile requires, whereas drug→gene response demands flexible composition and never enters the causally active subspace.

The bottleneck was the tokenisation, and it is repairable. Because a cell sentence ranks genes by expression, 83% of the target tokens are identical between any two drugs in the same context, so cross-entropy allocates roughly 1% of its gradient to drug identity. This explains a uniform pattern of failure: an entire ε -ladder of target constructions, from

a single state-matched treated cell to a full pseudobulk collapse, stays in the near-null regime, because all of them change *what the target is* and none change *what the tokens encode*. (One arm, the optimal-transport target, does show a weak effect once a sharper stratified test is applied — +0.0263 [+0.0069, +0.0476] — but roughly five-fold below what re-encoding achieves, which is the distinction that matters.) Re-encoding the target as the drug-specific residual raises the differing fraction to 59% and produces the first non-null drug effect in this project, with a monotone gradient across swap dissimilarity and a matched control that stays null.

And the repaired drug use generalises. On a holdout keyed at the (drug, cell line) level, pairings never seen in training score +0.1002 [+0.0661, +0.1368], indistinguishable from trained conditions. There is no memorisation premium: the model learned a drug representation that transfers to contexts in which it never observed that drug. Held-out *drugs* remain null, the control firing as designed.

But a lookup table wins — and it is not the ceiling either. A per-drug mean residual measured in other cell lines scores 0.963 against a 0.968 replicate ceiling; the model reaches 0.639. The natural reading of those three numbers is that the drug’s response is essentially a per-drug constant, that a conditional model therefore has almost nothing to condition on, and that the correct formulation of the task is retrieval rather than generation. We initially drew exactly that conclusion, and it is wrong.

It is wrong because NIR compresses near the top of its range — the same class of failure this thesis opens by documenting, arriving a second time in our own preferred instrument, at the ceiling instead of the floor. Measured in cosine space with measurement noise divided out, the transfer coefficient is $\mathcal{T}=0.557$ [0.513,0.601] against a simulated no-interaction null of 1.001, with a structure-matched negative control at +0.000. Roughly 45% of the drug-specific signal is a drug×cell-line interaction. The per-drug constant is what the lookup captures and what the model gropes toward; the interaction is reached by neither, and by nothing else in the literature we are aware of.

So the honest statement of what the model does is sharper, and less flattering, than “it misses the interaction”. It fails to recover the *main effect* that a numpy dictionary obtains for free, and the interaction is a further prize that no method here — ours least of all — comes near. Two different failures, and collapsing them undersells both.

That conclusion connects this work to an assumption the field has carried since the Connectivity Map: that a compound induces a characteristic signature stable enough across cellular contexts for signature-similarity querying to be meaningful. This thesis measures that assumption directly, with a calibrated metric and a proper replicate ceiling, at single-cell resolution — and the modern finding that a per-perturbation mean is hard to beat may be less a warning about weak baselines than a restatement of it.

It also places this work as the single-cell, generative instance of DrEval’s critique [13]. DrEval audits drug *sensitivity* prediction and finds that deep models barely outperform a naive additive predictor of mean drug and cell-line effects. Our `drug_lookup` is that predictor’s drug term, and the finding replicates on a 946-dimensional transcriptomic response: 0.963 against a 0.968 replicate ceiling, versus 0.639 for the model. Three of DrEval’s six pathologies — pseudoreplication, biased evaluation metrics, and missing ablation studies — turn out to structure the entire methodology of this thesis, which arrived at clustered bootstraps over cell lines, a discrimination metric immune to between-drug variance, and a token-level scramble ablation independently. That convergence is worth recording: the controls a critical benchmark demands are the same controls a single negative result needs in order to mean anything.

Where the opportunity actually is. Two further measurements locate it precisely, and between them they redirect what follows from this work.

The drug×cell-line interaction is large — roughly 45% of the drug-specific variance — but it has *no learnable structure*: no cell line modulates different drugs in a consistent direction, with an excess within-line consistency of -0.007 and, at plate scope, an interval excluding anything above $+0.002$. So the interaction behaves as irreducible noise for any predictor, and a lookup scoring 0.963 against a 0.968 ceiling is not evidence that the target has vanished but that the lookup is already at the *achievable* ceiling. On seen drugs, nothing beats retrieval, and that is a statement about the data rather than about any model.

But a lookup requires the drug to have been seen. For drugs that have not been, protein-target and mechanism annotations carry real signal against count-matched nulls ($+0.084$ [$+0.059$, $+0.110$] and $+0.078$ [$+0.056$, $+0.103$]), while chemical structure does not. This overturns a pre-registration made earlier in this project, which inferred that unseen drugs were unreachable from a gate that tested chemical *structure* — the channel our own drug-knowledge specification ranked lowest — and never read the target annotations sitting in the atlas’s own metadata.

So the thesis does not end in closure. It ends with the opportunity located: on seen drugs a generative model has nothing to add over retrieval, because the residual part is unpredictable; on unseen drugs, where retrieval is impossible by construction, drug-side knowledge works and a model can do something a dictionary cannot. That is also the setting the benchmark literature calls Leave-Drugs-Out, and the one that matters for drug design.

A note on method. The substantive results here rest on controls that were added in response to specific objections: plate-matched comparison sets after a zero-information baseline more than doubled its score when the plate was free to vary; a swap-distance

sweep after it was pointed out that scrambling to a similar drug is a weak test; a removal gate before trusting any ablation null; matched-norm injection before reading a swap; clustered bootstraps over cell lines rather than conditions; and a pre-registered interpretation written before the decisive result, whose middle branch is the one that fired. Several claims were retracted along the way and are struck rather than deleted. In a literature where the central problem is that uninformative baselines score well, the controls are not an appendix to the result — they are most of it.

References

- [1] Jesse Zhang et al. “Tahoe-100M: A Giga-Scale Single-Cell Perturbation Atlas for Context-Dependent Gene Function and Cellular Modeling”. In: *bioRxiv* (2025). Preprint, posted February 2025 (v3, 10 May 2025); not peer reviewed. Tahoe Therapeutics, formerly Vevo Therapeutics. Dataset: <https://huggingface.co/datasets/tahoebio/Tahoe-100M>. DOI: 10.1101/2025.02.20.639398.
- [2] Daniel Levine et al. “Cell2Sentence: Teaching Large Language Models the Language of Biology”. In: *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Vol. 235. Proceedings of Machine Learning Research. Preprint: bioRxiv 2023.09.11.557287. PMLR, 2024, pp. 27299–27325.
- [3] Syed Asad Rizvi et al. “Scaling Large Language Models for Next-Generation Single-Cell Analysis”. In: *bioRxiv* (2025). Preprint, v1 posted 17 April 2025; author list above is v4, posted 1 January 2026. Not peer reviewed. DOI: 10.1101/2025.04.14.648850.
- [4] Haotian Cui et al. “scGPT: toward building a foundation model for single-cell multi-omics using generative AI”. In: *Nature Methods* 21.8 (2024), pp. 1470–1480. DOI: 10.1038/s41592-024-02201-0.
- [5] Christina V. Theodoris et al. “Transfer learning enables predictions in network biology”. In: *Nature* 618.7965 (2023), pp. 616–624. DOI: 10.1038/s41586-023-06139-9.
- [6] Charlotte Bunne et al. “How to build the virtual cell with artificial intelligence: Priorities and opportunities”. In: *Cell* 187.25 (2024), pp. 7045–7063. DOI: 10.1016/j.cell.2024.11.015.
- [7] Stella Biderman et al. “Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling”. In: *Proceedings of the 40th International Conference on Machine Learning*. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 2397–2430. URL: <https://proceedings.mlr.press/v202/biderman23a.html>.
- [8] Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. “Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines”. In: *Nature Methods* 22.8 (2025), pp. 1657–1661. DOI: 10.1038/s41592-025-02772-6.
- [9] Gerold Csentes et al. “Benchmarking foundation cell models for post-perturbation RNA-seq prediction”. In: *BMC Genomics* 26.1 (2025), p. 393. DOI: 10.1186/s12864-025-11600-2.

- [10] Ihab Bendidi et al. “Benchmarking transcriptomics foundation models for perturbation analysis: one PCA still rules them all”. In: *arXiv preprint arXiv:2410.13956* (2024).
- [11] Eric Kernfeld et al. “A comparison of computational methods for expression forecasting”. In: *Genome Biology* 26.1 (2025). Preprint: bioRxiv 2023.07.28.551039, titled “A systematic comparison of computational methods for expression forecasting”, p. 388. DOI: 10.1186/s13059-025-03840-y.
- [12] Tony Hauptmann and Stefan Kramer. “A fair experimental comparison of neural network architectures for latent representations of multi-omics for drug response prediction”. In: *BMC Bioinformatics* 24.1 (2023), p. 45. DOI: 10.1186/s12859-023-05166-7.
- [13] Judith Bernett et al. “From Hype to Health Check: Critical Evaluation of Drug Response Prediction Models with DrEval”. In: *bioRxiv* (2025). Preprint, posted 29 May 2025. DOI: 10.1101/2025.05.26.655288.
- [14] Henry E. Miller et al. “Deep Learning-Based Genetic Perturbation Models Do Outperform Uninformative Baselines on Well-Calibrated Metrics”. In: *bioRxiv* (2025). Preprint, posted 21 October 2025. Shift Bioscience Ltd. DOI: 10.1101/2025.10.20.683304.
- [15] Gabriel M. Mejia et al. “Diversity by Design: Addressing Mode Collapse Improves scRNA-seq Perturbation Modeling on Well-Calibrated Metrics”. In: *arXiv preprint arXiv:2506.22641* (2025).
- [16] Yan Wu et al. “PerturBench: Benchmarking Machine Learning Models for Cellular Perturbation Analysis”. In: *arXiv preprint arXiv:2408.10609* (2024).
- [17] Stuart H. Hurlbert. “Pseudoreplication and the Design of Ecological Field Experiments”. In: *Ecological Monographs* 54.2 (1984), pp. 187–211. DOI: 10.2307/1942661.
- [18] Alexander Partin et al. “Deep learning methods for drug response prediction in cancer: Predominant and emerging trends”. In: *Frontiers in Medicine* 10 (2023). DOI: 10.3389/fmed.2023.1086097.
- [19] Justin Lamb et al. “The Connectivity Map: using gene-expression signatures to connect small molecules, genes, and disease”. In: *Science* 313.5795 (2006), pp. 1929–1935. DOI: 10.1126/science.1132939.
- [20] Aravind Subramanian et al. “A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles”. In: *Cell* 171.6 (2017), 1437–1452.e17. DOI: 10.1016/j.cell.2017.10.049.

- [21] Yanai Elazar et al. “Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals”. In: *Transactions of the Association for Computational Linguistics* 9 (2021), pp. 160–175. DOI: 10.1162/tac1_a_00359.
- [22] Wes Gurnee, Nicholas Sofroniew, Adam Pearce, et al. “Verbalizable Representations Form a Global Workspace in Language Models”. In: *Transformer Circuits Thread* (2026). Published 6 July 2026. <https://transformer-circuits.pub/2026/workspace/index.html>.
- [23] Nelson Elhage et al. “Toy Models of Superposition”. In: *Transformer Circuits Thread* (2022). arXiv:2209.10652.
- [24] Marco Cuturi. “Sinkhorn Distances: Lightspeed Computation of Optimal Transport”. In: *Advances in Neural Information Processing Systems 26 (NIPS 2013)*. Ed. by C.J. Burges et al. Vol. 26. Curran Associates, Inc., 2013, pp. 2292–2300. URL: https://proceedings.neurips.cc/paper_files/paper/2013/file/af21d0c97db2e27e13572cbf59eb343d-Paper.pdf.
- [25] Geoffrey Schiebinger et al. “Optimal-Transport Analysis of Single-Cell Gene Expression Identifies Developmental Trajectories in Reprogramming”. In: *Cell* 176.4 (2019), 928–943.e22. DOI: 10.1016/j.cell.2019.01.006.
- [26] Romain Lopez et al. “Deep generative modeling for single-cell transcriptomics”. In: *Nature Methods* 15.12 (2018), pp. 1053–1058. DOI: 10.1038/s41592-018-0229-2.

A Appendix

A.1 Estimator validation against planted ground truth

Three of the measurements in this thesis rest on estimators built for the purpose rather than taken off the shelf. Each was validated by planting a known answer and checking recovery — in both directions, since an estimator that only ever returns the answer one hopes for is not evidence.

Estimator	Planted truth	Recovered
transfer coefficient $\mathcal{T}$	$\mathcal{T} = 1.00$ (no interaction)	0.999
	$\mathcal{T} = 0.70$	0.703
	$\mathcal{T} = 0.40$	0.404
channel gate	a channel that genuinely predicts	+0.397 (sd 0.029) over 24 worlds
	a channel carrying nothing	−0.016 (sd 0.091) over 24 worlds
κ -structure	a consistent per-line modulation	excess +1.085
	an idiosyncratic interaction	excess +0.001

Table 21: All three recover planted answers in both directions.

Two of these validations changed the estimators rather than merely endorsing them. The channel gate’s count-matched null originally drew its random partners from a pool *excluding* the channel’s own selections, which depletes the null of exactly the partners the channel chose and anti-correlates the two arms. And the useless-channel gap has a standard deviation of roughly 0.09 on a single small world, so the original validation — asserting that one draw fell within 0.10 — tested luck rather than correctness; it now averages 24 independent worlds. That variance is also why the reported channel results are gated on a clustered interval rather than on a point estimate.

A.2 Absent-gene convention

A predicted cell sentence lists only expressed genes, so some panel genes are unranked and a convention is required. Two were carried throughout: **worst** assigns absent genes the last rank P , **francesca** a fixed middle rank $P/2$. The choice does not matter — spike-in DE- Δr reads 0.952 under the first and 0.954 under the second, and no ceiling or baseline in this thesis moves by more than 0.01 between them. All figures in the body are quoted under **worst**.

A.3 Swap-distance sweep

The stratified scramble of Section 3.4 was preceded by a continuous version: Δ_{scr} measured as a function of the response distance between the real and the swapped-in drug, binned

into quintiles.

Bin (swap distance)	single-cell model	optimal-transport model
0 — nearest (2.75–4.03)	−0.046	−0.001
1 (4.03–4.60)	+0.060	+0.031
2 (4.60–4.93)	−0.006	+0.017
3 (4.93–5.32)	−0.004	+0.043
4 — farthest (5.32–6.25)	−0.019	−0.005
trend, corr(distance, Δ_{scr})	+0.051	+0.012

Table 22: No bin in either model has an interval excluding zero, and the trend correlations are ≈ 0 . 200 (drug, swap) pairs over 20 cell lines, $n = 40$ per bin.

Two features make this an argument rather than five nulls. The single borderline value sits in a *middle* bin, whereas a genuine dissimilarity effect would be monotone and peak in the far bin. And the comparison is paired: the model’s own NIR drifts across bins from 0.497 to 0.326, because drugs that are far from everything populate the far bins, but Δ_{scr} uses the same drug on both arms so that drift cancels.

The intervals are wide — $n = 40$ per bin gives half-widths of 0.06–0.10 — so this sweep excludes a large effect rather than establishing a null, which is why the stratified scramble replaced it in the body.

A.4 Training configuration

Every arm was cold-started from the same base checkpoint with identical hyperparameters, so that the target or objective is the only variable. One exception is material and is stated wherever the arm is quoted.

Arm	Target	Training	Note
single cell	the observed treated cell	1 epoch	
consensus	group pseudobulk	$\approx 60\%$ of an epoch	tier-1 NIR unscored
optimal transport (T2)	barycentric, $\varepsilon = 0.5$	1 epoch, 23,842 steps	
residual	signed residual signature	1 epoch, 15,340 steps	
residual (holdout)	as above, 738 conditions withheld	1 epoch	

Table 23: Learning rate 10^{-5} , batch size 1, gradient accumulation 16, maximum length 8192, bf16, weight decay 0.01, warmup ratio 0.03 throughout. The consensus arm stopped early on a flat loss; since it is one endpoint of the ε -ladder, the ladder’s negative rests on the other two.

A.5 Calibration self-test

The calibration framework of Section 3.3 is itself checked, since a metric audit whose instrument is untested proves nothing. Beyond deterministic unit checks — a perfect prediction must reach the metric’s maximum, a zero shift and a flat truth must return no score — the self-test asserts the exploit under a *second, independent* zero-information baseline: a leave-one-out mean scores $\text{DE-}\Delta r = 1.000$ while scoring $\text{NIR} = 0.000$. Two unrelated constructions both saturating the standard metric, and both scoring zero on the discrimination metric, establishes the exploit as a property of the metric rather than of any one baseline.